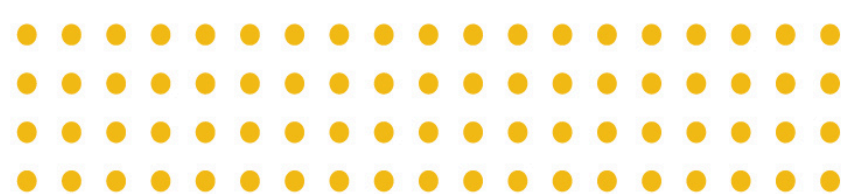


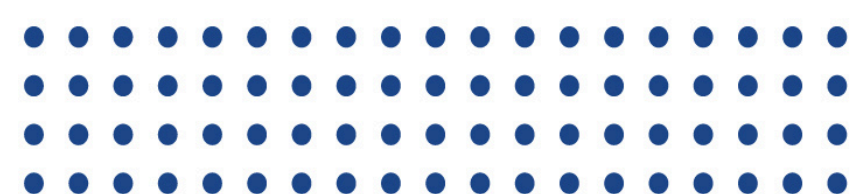


Mahavir Education Trust's
Shah & Anchor Kutchhi Engineering College
An Autonomous Institute Affiliated to University of Mumbai



2025-26

Scheme & Syllabus



Bachelor of Technology (B.Tech.) **in** **Information Technology**

Final Year B.Tech.
(with effect from 2025-2026)



Mahavir Education Trust's

Shah & Anchor Kutchhi Engineering College

An Autonomous Institute Affiliated to University of Mumbai

CURRICULUM STRUCTURE FINAL YEAR B.Tech.

Information Technology

REVISION: R1-V0-2025-26

Effective for Academic Year 2025-26 and 2026-27 only

Approved by Board of Studies on 17/06/2025

Approved by Academic Council on 19/06/2025


Dr. Swati Nadkarni

I/C HOD

Chairman BoS

Department of Information Technology
Shah & Anchor Kutchhi Engineering College
Chembur, Mumbai - 400088




Dr. Bhavesh Patel

Principal

Dr. Bhavesh Patel

Chairman

Academic Council

Shah & Anchor Kutchhi Engineering College
Chembur, Mumbai - 400088



Mahavir Education Trust's

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An Autonomous Institute Affiliated to University of Mumbai

UG Program in Information Technology

VISION:

To develop skilled Information Technology engineers by offering quality education with rapidly evolving advanced technologies to embark upon global challenges and standards for the betterment of society.

MISSION:

1. To create competent and trained professionals in Information Technology who shall contribute towards the advancement of engineering, science and technology useful for the society.
2. To impart quality and value based education to raise the satisfaction level of all stakeholders.
3. To generate technically sound professionals and entrepreneurs to become part of the industry and research organizations at national levels.

PROGRAM SPECIFIC OUTCOMES (PSO)

- **PSO1:** The Information Technology graduates are able to analyze, design, develop, test and apply management principles, mathematical foundations in the development of IT based solutions for real world and open-ended problems.
- **PSO2:** The Information Technology graduates are able to perform various roles in creating innovative career paths: to be an entrepreneur, a successful professional, pursue higher studies with realization of moral values and ethics.

PROGRAM EDUCATIONAL OBJECTIVES (PEO)

- **PEO1:** A graduate will excel in a professional career and contribute to social needs through Information Technology.
- **PEO2:** A graduate will be integrated into the world of practicing professionals for collaborations, pursue higher education, conduct research, demonstrate professionalism and ethics.
- **PEO3:** A graduate will Exhibit innovation, team work, leadership and communication skills through lifelong learning.



UG Program in Information Technology

KNOWLEDGE AND ATTITUDE PROFILE (WK)

1. **WK1:** A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.
2. **WK2:** Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.
3. **WK3:** A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
4. **WK4:** Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.
5. **WK5:** Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.
6. **WK6:** Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.
7. **WK7:** Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.
8. **WK8:** Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.
9. **WK9:** Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.



UG Program in Information Technology

PROGRAM OUTCOMES (PO)

1. **PO1: Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
2. **PO2: Problem Analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
3. **PO3: Design/Development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
4. **PO4: Conduct Investigations of Complex Problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis interpretation of data to provide valid conclusions. (WK8).
5. **PO5: Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
6. **PO6: The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
7. **PO7: Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national international laws. (WK9)
8. **PO8: Individual and Collaborative Team work:** Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
9. **PO9: Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.
10. **PO10: Project Management and Finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
11. **PO11: Life-Long Learning:** Recognize the need for, and have the preparation and ability for
i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)



Mahavir Education Trust's

Shah & Anchor Kutchhi Engineering College

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UG Program in Information Technology

PREAMBLE

Shah and Anchor Kutchhi Engineering College (SAKEC) has been granted academic autonomy by University Grants Commission (UGC) from Academic Year 2024-25 for 10 years. The program in Information Technology (IT) has prepared an Autonomy Scheme curriculum for 4 years undergraduate (UG) and 2 years of post-graduate (PG). The overall credits and courses for the first year of engineering and second year of engineering curriculum is in line with the NEP 2020. The first year of engineering consists of foundational courses in basic science and engineering science category and other categories are value education, ability enhancement, and co-curricular courses. Second year of engineering contains mainly program core courses. Multidisciplinary courses and entrepreneurship category courses are introduced for the second year. The emphasis is on vocational and skill development courses for all four years of engineering. The first year and second year focuses on building the foundations, and is structured as per NEP credit distribution to categories. The third year and final year is for developing the skills and knowledge of the students in various domains through electives. The curriculum is designed to cater to industry requirements and different students may follow different paths and take different set of courses in it. The curriculum is designed to impart technical knowledge as per latest industry standards. In order to gain practical skills learners are offered various skill based courses, program electives, project based learning. New courses are introduced to cater to co-curricular and extra curricular development of learners. It provides enhanced learning experience with the introduction of activity based learning and project based learning. Human values are instilled in the learners by incorporating social involvement and real world problem solving through field visits, case studies, project developments and internships. Therefore, it boosts the learner's profile for taking up higher studies as well as for finding appropriate jobs. It has also given ample opportunities to explore entrepreneurship as a career option. It also focuses on continuous internal assessment which is very encouraging and beneficial for the learners. The factors which led to this present curriculum development are research, innovation and outcome based education. The shift in focus from teacher-centric to learner-centric education for designing this curriculum will definitely improve quality of education. This curriculum is an amalgamation of knowledge, skills and value education. The motto of the program in information technology is Technology driven holistic development through the right blend of courses.



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UG Program in Information Technology

Abbreviations used in course codes for course categories based on NEP

Category	Course Category	Abbreviation
BSC/ESC	Basic Science Courses	BS
	Engineering Science Courses	ES
Program Courses	Programme Core Course	PC
	Programme Elective Course	PE
Multidisciplinary Courses	Multidisciplinary Minor	MD
	Open Elective	OE
Skill Courses	Vocational & Skill Enhancement Course	VS
Humanities Social Science & Management	Entrepreneurship / Economics / Management Courses	ED
	Indian Knowledge System	IK
	Value Education Course	VE
	Ability Enhancement	AE
Experiential Learning Courses	Research Methodology	RM
	Field Project/Community Engagement Project	CP
	Project	PR
	Internship	IN
Liberal Learning Courses	Co-curricular course	CC



Mahavir Education Trust's

Shah & Anchor Kutchhi Engineering College

An Autonomous Institute Affiliated to University of Mumbai

Program Structure for Final Year in B.tech in Information Technology (with effect from 2025–2026)

Semester VII

			Learning Scheme					Credits	Assessment Scheme																								
Sr. No.	Course Title	Course Code	Actual Contact Hrs/Week			SL	Total		Exam	Theory					Practical				SLA		Total												
										Duration	CIA		ESE	Total		CIAP	ESEP	Total															
			CL	TL	LL						Notional	in Hrs										MSE	CCE	Max	Max	Max	Max	Min	Max	Max	Max	Min	Max
			1	Artificial Intelligence and Data Science –II	ITCR1701						3	1								2		2	120	4	2+2	20	20	60	100	40	20	30	50
2	Internet of Everything	ITCR1702	3	1	2	2	120		4	2+2	20	20	60	100	40	20	30	50	20	25	10	175											
3	Department Level Optional Course-3	ITDLOC1703X	3	1	-	2	90	3	2	20	20	60	100	40	-	-	-	-	25	10	125												
4	Department Level Optional Course-4	ITDLOC1704X	2	-	-	2	60	2	-	-	50	-	50	20	-	-	-	-	25	10	75												
5	Institute Level Optional Course-1	XXILOCR1705	3	1	-	2	90	3	2	20	20	60	100	40	-	-	-	-	25	10	125												
6	Secure Application Development Lab	ITLR1706	-	-	2	-	30	1	2	-	-	-	-	-	20	30	50	20	-	-	50												
7	Major Project -I	ITLR1707	-	-	6	-	90	3	2	-	-	-	-	-	25	25	50	20	-	-	50												
Total			14	4	12	10	600	20	-	80	130	240	450	-	85	115	200	-	125	-	775												

Abbreviations:

CL: Classroom Learning TL: Tutorial Learning LL: Laboratory Learning SL: Self Learning (Activity/Assignment/Micro-Project) CIA: Continuous Internal Assessment
CCE: Continuous Comprehensive Evaluation MSE: Mid Semester Examination ESE: End Semester Examination CIAP: Continuous Internal Assessment Practical ESEP: End Semester Examination Practical
SLA: Self Learning Assessment

Note:

1. Notional Learning hours for the semester are (CL+LL+TL+SL) hrs × 15 Weeks.
2. 1 credit is equivalent to 30 Notional hrs.
3. Self learning hours shall not be reflected in the Time Table.
4. SLA will have separate head of passing



**Program Structure for Final Year B.Tech.in
Information Technology
(with effect from 2025-2026)**

Department Level Optional Course -3

Sr. No	Course Code	Course Name
1	ITDLOCR17031	Storage Area Network
2	ITDLOCR17032	High Performance Computing
3	ITDLOCR17033	Infrastructure Security
4	ITDLOCR17034	Software Testing & Quality Assurance

Department Level Optional Course -4

Sr. No	Course Code	Course Name
1	ITDLOCR17041	Mobile Adhoc Network
2	ITDLOCR17042	Augmented & Virtual Reality
3	ITDLOCR17043	Quantum Computing
4	ITDLOCR17044	Information Retrieval System

Institute Level Optional Course -1

Sr. No	Course Code	Course Name
1	ECILOCR1705	Management Information System
2	CSILOCR1705	Research Methodology
3	ETILOCR1705	Disaster Management and Mitigation Measures
4	ADILOCR1705	Entrepreneurship Development and Management



Program Structure for Final Year B.Tech. in Information Technology (with effect from 2025–2026)

Semester VIII

			Learning Scheme					Credits	Assessment Scheme													
Sr. No.	Course Title	Course Code	Actual Contact Hrs/Week			SL	Total		Exam	Theory					Practical				SLA		Total	
			CL	TL	LL					Duration in Hrs	CIA		ESE	Total		CIAP	ESEP	Total				
											MSE	CCE		Max	Min			Max				Min
											Max	Max		Max	Max			Min	Max	Min		
1	Blockchain and DLT	ITCR1801	3	1	2	2	120	4	2+2	20	20	60	100	40	10	15	25	10	25	10	150	
2	Department Level Optional Course- 5	ITDLOC1802X	3	1	-	2	90	3	2	20	20	60	100	40	-	-	-	-	25	10	125	
3	Department Level Optional Course-6	ITDLOC1803X	3	1	-	2	90	3	2	20	20	60	100	40	-	-	-	-	25	10	125	
4	Institute Level Optional Course-2	XXILOCR1804	3	1	-	2	90	3	2	20	20	60	100	40	-	-	-	-	25	10	125	
5	Cloud Computing Lab	ITLR1805	-	-	2	-	30	1	2	-	-	-	-	-	20	30	50	20	-	-	50	
6	Major Project -II	ITLR1806	-	-	12	-	180	6	2	-	-	-	-	-	100	50	150	60	-	-	150	
Total			12	4	16	8	600	20	-	80	80	240	400	-	130	95	225	-	100	-	725	

Abbreviations:

CL: Classroom Learning TL: Tutorial Learning LL: Laboratory Learning SL: Self Learning (Activity/Assignment/Micro-Project) CIA: Continuous Internal Assessment
 CCE: Continuous Comprehensive Evaluation MSE: Mid Semester Examination ESE: End Semester Examination CIAP: Continuous Internal Assessment Practical ESEP: End Semester Examination Practical
 SLA: Self Learning Assessment

Note:

1. Notional Learning hours for the semester are (CL+LL+TL+SL) hrs × 15 Weeks.
2. 1 credit is equivalent to 30 Notional hrs.
3. Self learning hours shall not be reflected in the Time Table.
4. SLA will have separate head of passing



**Program Structure for Final Year B.Tech.in
Information Technology
(with effect from 2025-2026)**

Department Level Optional Course -5

Sr. No	Course Code	Course Name
1	ITDLOCR18031	Big Data Analytics
2	ITDLOCR18022	Reinforcement Learning
3	ITDLOCR18023	Simulation and Modelling
4	ITDLOCR18024	Knowledge Management

Department Level Optional Course -6

Sr. No	Course Code	Course Name
1	ITDLOCR18031	User Interface Design
2	ITDLOCR18032	Robotics
3	ITDLOCR18033	Enterprise Resource Planning
4	ITDLOCR18034	CloudComputing and Services

Institute Level Optional Course -2

Sr. No	Course Code	Course Name
1	CMILOCR1804	Project Management
2	ITILOCR1804	Finance Management
3	ECILOCR1804	Environmental Management
4	CSILOCR1804	IPR and Patenting



Mahavir Education Trust's

Shah & Anchor Kutchhi Engineering College

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Semester VII



Program	Final Year B.Tech. Information Technology (Semester VII)	CL	TL	LL	SL	C
Course Name: Artificial Intelligence and Data Science –II	Course Code: ITCR1701	3	1	2	2	4
Course Type:	Programme Core (PC)					
Pre-requisite:	Data Structure, Database Management System.					

I RATIONALE

In today's era of digital transformation, Artificial Intelligence (AI) and Data Science (DS) play a critical role across diverse domains. From healthcare and finance to manufacturing, retail, education, and governance, AI and DS are driving innovation, improving decision-making, and automating complex processes.

II COMPETENCY

Demonstrate the ability for design, implement, and evaluate advanced artificial intelligence and data science models, including generative models, reinforcement learning, and multi-agent systems, while addressing real-world problems with ethical and strategic decision-making.

III COURSE OUTCOMES (COs)

After the completion of course based learning Students will be able to,

Course Outcome Number	Course Outcome Statement
1	Demonstrate the uncertainty process and cognitive computations and applications.
2	Apply the concept of Fuzzy Logic and applications.
3	Illustrate Deep Learning techniques.
4	Analyze Machine Learning classification techniques for real world applications.

IV TEACHING-LEARNING & ASSESSMENT SCHEME

				Learning Scheme				Credits	Assessment Scheme																	
Sr. No.	Course Title	Course Type	Course Code	Actual Contact Hrs/Week			SL		Total		Theory					Practical					SLA					Total
				CL	TL	LL				Exam	CIA		ESE	Total	CIAP	ESEP	Total									
											Duration in Hrs	Max						Max	Max	Max	Min	Max	Max	Max	Min	
1	Artificial Intelligence and Data Science – II	PC	ITCR1701	3	1	2	2	120	4	2+2	20	20	20	60	100	40	20	30	50	20	25	10	175			

V LABORATORY LEARNING OUTCOME AND SUGGESTED LIST OF EXPERIMENTS (Minimum 8)

Sr. No.	Laboratory Learning Outcome	Laboratory Experiment Titles	Relevant COs
1	Implement Bayesian Network models on a dataset using Python.	Implementation of different operation on a dataset by Bayesian Network in Python.	CO1
2	Implement data cleaning operations on a dataset used in cognitive applications.	Implementation of data cleaning operations on a dataset in cognitive application.	CO1
3	Implement Matplotlib and Seaborn libraries in Python.	Implementation of Matplotlib and Seaborn in Python.	CO2
4	Implement slicing, indexing, and data manipulation using fuzzy logic to clean and preprocess.	Implementation of Slicing, Indexing, Manipulating and Cleaning with using fuzzy logic.	CO2
5	Implement the Python libraries and data structures to implement RNN or LSTM models for sequential data processing and prediction.	Implementation of Python Libraries and Data Structure by using RNN or LSTM.	CO3
6	Implement data visualization and exploratory data analysis (EDA) tools to preprocess and interpret datasets. used for training ANN or CNN.	Implementation of data visualization/ exploratory data analysis tools by using ANN or CNN.	CO3
7	Implement the concept of arrays, Pandas, and CSV file operations in Python to preprocess and manage data.	Implementation of Arrays, Pandas and CSV files for some Applications (Ane one) Handwritten Digit Recognition System, Image Classification, Image Caption Generator.	CO3
8	Implement predictive models using supervised learning algorithms such as AdaBoost and Random Forests in Python.	Implementation of Predictive Model using Python, Ada-Boosting, Random forests And show Evaluation of Classification Algorithms.	CO4
9	Implementation of linear regression using Python to model the relationship between variables	Implementation of Linear Regression using Python.	CO4
10	Implement logistic regression using Python to perform binary classification tasks and evaluate.	Implementation of Logistic Regression using Python.	CO4

VI THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
1	CO1	1. Apply the concept of uncertainty in Artificial Intelligence by identifying and addressing its various sources in decision-making processes. 2. Apply the structure and semantics of Bayesian networks to represent probabilistic relationships using nodes, edges, and conditional probability tables. 3. Illustrate the role of HMM and GMM components in building AI systems for pattern recognition. 4. Demonstrate foundational cognitive computing concepts by applying perception, reasoning, learning, and natural interaction to NLP-enabled intelligent systems. 5. Apply advanced analytics techniques such as machine learning, natural language processing, and predictive modeling to enhance the capabilities of cognitive computing systems.	Uncertainty and Cognitive Computing 1.1 Uncertainty in AI, Inference using full joint distributions. 1.2 Bayes Theorem, the semantics of Bayesian Networks, Inference in Bayesian networks, decision theory, and Markov decision processes. 1.3 Hidden Markov Model (HMM), Gaussian Mixture Model (GMM). 1.4 Foundation of Cognitive Computing, Design Principles for Cognitive Systems, Natural Language Processing in Support of a Cognitive System. 1.5 Applying Advanced Analytics to Cognitive Computing, The Process of Building a Cognitive Application, Representing Knowledge in Taxonomies and Ontologies.	10

(Continued on next page)

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
2	CO2	1. Demonstrate the application of fuzzy set principles by differentiating them from crisp sets in decision-making scenarios. 2. Implement defuzzification methods within fuzzy inference systems to generate precise, crisp outputs. 3. Use fuzzy logic principles to develop control strategies for appliances that operate under uncertain or vague conditions. 4. Apply data science concepts to real-world case studies from domains such as healthcare, finance, retail, and social media.	Fuzzy Logic and Its Applications Trends and applications in Data Science 2.1 Introduction to Fuzzy Sets, Properties of Fuzzy Sets, Operations on Fuzzy Sets, Fuzzy Membership Functions, Fuzzy Relations with Operations and its Properties, Fuzzy Composition: Max-Min Composition, Max-Product Composition. 2.2 Defuzzification Methods, Architecture of Mamdani Type Fuzzy Control System. 2.3 Design of Fuzzy Controllers like Domestic Shower Controller, Washing Machine Controller, Water Purifier Controller, etc. 2.4 Data Science: applications and case studies, Data science for text, image, video, audio. Data science for Multi modal applications.	12
3	CO3	1. Illustrate the working of deep learning systems through application of their architecture, layered design, and training strategies. 2. Demonstrate the process of training deep learning models by applying forward propagation, loss functions, and backpropagation. 3. Demonstrate the use of CNNs in solving domain-specific challenges related to visual perception and pattern recognition. 4. Apply the concept and architecture of autoencoders in unsupervised deep learning for feature extraction and dimensionality reduction.	Introduction to Deep Learning 3.1 Introduction to Deep Learning, ANN, Machine Learning Vs Deep Learning. 3.2 Working of Deep Learning; Convolution Neural Network: Introduction, Components of CNN Architecture, Properties of CNN, Architectures of CNN. 3.3 Applications of CNN, Recurrent Neural Network: Introduction, Simple RNN, LSTM, Implementation, Deep RNN. 3.4 Auto encoder: Introduction, Features, Types, Applications of Deep Learning. Introduction to Reinforcement Learning.	12

(Continued on next page)

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
4	CO4	1. Use ensemble techniques such as bagging, boosting, and stacking to enhance model accuracy and stability. 2. Implement evaluation techniques using key metrics like precision, recall, and ROC-AUC to measure classifier accuracy. 3. Illustrate the significance of classification outcomes by applying cost-benefit reasoning in decision-making contexts.	Advanced ML Classification Techniques 4.1 Ensemble Classifiers: Introduction to Ensemble Methods, Bagging, Boosting, Random forests, Improving classification accuracy of Class-Imbalanced Data. 4.2 Metrics for Evaluating Classifier Performance, Holdout Method and Random Sub-sampling, Cross-Validation, Bootstrap, Model Selection Using Statistical Tests of Significance. 4.3 Comparing Classifiers Based on Cost-Benefit and ROC Curves.	11

VII SUGGESTED SELF LEARNING ACTIVITIES

1. Explore tools like Weka or RapidMiner or Orange for classification model evaluation.
2. Make a comparative table showing performance metrics for different ensemble techniques.
3. Create a summary on how deep learning is used in healthcare/finance/retail.

VIII SUGGESTED WEIGHTAGE TO ASSESSMENT PURPOSE

CO No.	Course Outcome	BL1	BL2	BL3	BL4	BL5	BL6	Total
1	Demonstrate the uncertainty process and cognitive computations and applications.	10%	30%	60%	-	-	-	100%
2	Apply the concept of Fuzzy Logic and applications.	10%	30%	60%	-	-	-	100%
3	Illustrate Deep Learning techniques.	10%	30%	40%	20%	-	-	100%
4	Analyze Machine Learning classification techniques for real world applications.	10%	30%	40%	20%	-	-	100%

IX SUGGESTED LEARNING MATERIALS / TEXTBOOKS / REFERENCE BOOKS

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Introduction to Data Science Practical Approach with R and Python	1 st Edition	B. Uma Maheshwari, R. Sujatha	Wiley	2021
2	Deep Learning Using Python	1 st Edition	Dr. S Lovelyn Rose, Dr. L Ashok Kumar, Dr. D Karthika Renuka	Wiley	2020
3	Deep Learning with Python	1 st Edition	François Chollet	Manning Publications	2018
4	Artificial Intelligence: A Modern Approach	3 rd Edition	Stuart Russell and Peter Norvig	Pearson Education	2015

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	A First Course in Artificial Intelligence	1 st Edition	Khemani	McGraw Hill	2017
2	Introduction to Machine Learning	3 rd Edition	Ethem Alpaydin	PHI Learning Pvt. Ltd.	2015
3	Deep Learning Illustrated: A Visual, Interactive Guide to Artificial Intelligence	1 st Edition	Jon Krohn, Grant Beyleveld, AglaéBassens	Pearson Education	2019
4	Artificial Intelligence with Python	1 st Edition	Prateek Joshi	Packt Publishing	2017

Online References

Sr. No	Website Name
A	https://www.appliedaicourse.com/blog/uncertainty-in-ai/
B	https://www.geeksforgeeks.org/artificial-intelligence/representing-knowledge/ tps://www.datacamp.com/blog/what-is-cognitive-computing
C	https://www.simplilearn.com/tutorials/deep-learning-tutorial/introduction-to-deep-learning

Program	Final Year B.Tech. Information Technology (Semester VII)	CL	TL	LL	SL	C
Course Name: Internet of Everything	Course Code: ITCR0702	3	1	2	2	4
Pre-requisite:	Wireless Technology					

I RATIONALE

The Internet of Everything (IoE) represents the next evolution of the Internet, enabling intelligent connections between people, processes, data, and things. As the digital world expands, there is a growing need for professionals who can understand and implement IoE technologies across diverse sectors like smart cities, healthcare, agriculture, and manufacturing. This course is designed to bridge theoretical knowledge with practical skills in areas such as networking, sensor integration, cloud computing, and data analytics. It empowers students to build scalable, secure, and intelligent systems that drive automation and enhance decision-making. With emphasis on interdisciplinary learning, real-time applications, and ethical considerations, the course prepares learners to address complex challenges in a hyper-connected world and contribute meaningfully to technological and societal advancement.

II COMPETENCY

The course enables students to apply principles of connectivity, data integration, and automation to develop smart, context-aware systems. Learners will gain the ability to design secure and scalable IoE solutions that integrate people, processes, data, and things. They will also analyze the impact of IoE technologies across various domains while considering ethical, privacy, and sustainability concerns, fostering innovation in real-world applications.

III COURSE OUTCOMES (COs)

After the completion of course based learning Students will be able to,

Course Outcome Number	Course Outcome Statement
1	Design a conceptual model showing how IoE can optimize services in a smart environment.
2	Analyze access technologies and protocols.
3	Evaluate data analytics and various supporting services for optimizing real-time IoE decision-making.
4	Design IOT application for a chosen domain demonstrating integration of sensors, connectivity, and intelligence.

IV TEACHING-LEARNING & ASSESSMENT SCHEME

				Learning Scheme				Credits	Assessment Scheme																	
Sr. No.	Course Title	Course Type	Course Code	Actual Contact Hrs/Week			SL		Total	Exam	Theory					Practical				SLA		Total				
				CL	TL	LL					Duration in Hrs	CIA		ESE	Total	CIAP	ESEP	Total								
												Max	Max					Max	Min				Max	Min	Max	Min
												Max	Max					Max	Min				Max	Min	Max	Min
1	Internet of Everything	PC	ITCR1702	3	1	2	2	120	4	2+2	20	20	20	60	100	40	20	30	50	20	25	10	175			

**V LABORATORY LEARNING OUTCOME AND SUGGESTED LIST OF EXPERIMENTS**

Sr. No.	Laboratory Learning Outcome	Laboratory Experiment Titles	Relevant COs
1	Study the Architecture of IOT	Study the Architecture of IOT	CO1
2	Study different IOT boards, sensors and Actuators.	Study different IOT boards, sensors and Actuators.	CO1
3	3. Identify different communication technologies and protocols in IOT.	Identify different communication technologies and protocols in IOT.	CO2
4	Create an IoT network scenario using Cooja simulator.	Create an IoT network scenario using Cooja simulator.	CO3
5	Design home automation communication access points using Cisco home packet tracer simulator.	Design home automation communication access points using Cisco home packet tracer simulator.	CO3
6	Design MicroProject	Design MicroProject	CO4

VI THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
1	CO1	1. Discuss the fundamentals of IOT 2. Apply Smart Object 3. Explain Architecture 4. Analyze Applications, Data, Network of Smart Services. 5. Discuss IoT Data Management and Compute Stack 6. Design IOT Application	Introduction to IOT Architecture 1.1 Fundamentals of IOT: Introduction to IoT- Defining IoT, Characteristics of IoT, Conceptual Framework of IoT, Physical design of IoT, Logical design of IoT, Functional blocks of IoT 1.2 Smart Object :Definition, Characteristics and Trends 1.3 Architecture : The IoT World Forum (IoTWF) Standardized Architecture :Layer 1-7, IT and OT Responsibilities in the IoT Reference Model, Additional IoT Reference Models A Simplified IoT Architecture ,The Core IoT Functional Stack ::Layer 1-3 1.4 Analytics Versus Control Applications, Data Versus Network Analytics Data Analytics Versus Business Benefits, Smart Services. 1.5 IoT Data Management and Compute Stack :Fog Computing , Edge Computing ,The Hierarchy of Edge, Fog, and Cloud	11
2	CO2	1. Examine IoT Access Technologies 2. Explain Network Layer 3. Examine Application Layer Protocols	IOT Access Technologies Protocols 2.1 IoT Access Technologies :RFID and NFC (Near-Field Communication), Physical and MAC layers, topology and Security of IEEE 802.15.4, 802.11ah and Lora WAN 2.2 Network Layer: IP versions, Constrained Nodes and Constrained Networks, 6LoWPAN 2.3 Application Layer Protocols: CoAP and MQTT, STOMP, AMQP, Comparison of Protocols.	12

(Continued on next page)

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
3	CO3	1. Evaluate Data Analytics 2. Explain Supporting Services	Data Analytics and Supporting Services 3.1 Data Analytics: Introduction, Structured Versus Unstructured Data, Data in Motion versus Data at Rest, IoT Data Analytics Challenges, Data Acquiring, Organizing in IoT/M2M 3.2 Supporting Services: Computing Using a Cloud Platform for IoT/M2M Applications/Services, Everything as a service and Cloud Service Models	12
4	CO4	1. Design Case studies 2. Discuss introduction to I-IoT and its Use cases 3. Differentiate IoT and I-IoT	IOT Applications 4.1 Case study related to: Home Automation (Smart lighting, Home intrusion detection), Cities (Smart Parking), Environment (Weather monitoring, weather reporting Bot, Air pollution monitoring, Forest fire detection, Agriculture (Smart irrigation), Smart Library. 4.2 Introduction to I-IoT, Use cases of the I-IoT, IoT and I-IoT – similarities and differences.	10

VII SUGGESTED SELF LEARNING ACTIVITIES

- * Explore the tools for IOT application development
- * Explore the protocols, standards required IOT application development
- * Explore for data analytics tools

Suggested Microprojects:

- * Automatic Street Light- Street Light should automatically ON at evening and automatically OFF at morning. LCD and Serial Monitor shows Light Intensity value on First Line and Status of Street Light on Second Line. USE RGB LED for street Light and use orange color.
- * Home Automation through PC- Design and develop project to control 8 home devices through PC serial monitor, LCD connected on project will shows Status of Devices is on or off. Also show the status of all devices on serial monitor.
- * Motion enabled Room Light- Light present in Room should automatically ON when human motion is detected and automatically OFF in the absence of human motion. LCD and Serial monitor shows appropriate message as “Motion detected! Light ON” and “No Motion! Light OFF” when particular condition fulfilled.
- * Electronic Smart Blind Stick- If someone is in front of blind person, LED and Buzzer should on and LCD will show the message” Obstacle. Be Alert” otherwise LED and Buzzer will remains off and LCD show the message “Safe.. Keep Walking”.
- * Electronic Notice Board- Any Message send from Serial Monitor should get displayed on LCD. When new message sends, previous message gets automatically erased and replaced with new message.

Teacher shall allocate any other microproject relevant to COs.

VIII SUGGESTED WEIGHTAGE TO ASSESSMENT PURPOSE

CO No.	Course Outcome	BL1	BL2	BL3	BL4	BL5	BL6	Total
1	Design a conceptual model showing how IoE can optimize services in a smart environment	10%	20%	20%	20%	10%	20%	100%
2	Analyze access technologies and protocols	10%	30%	30%	30%	-	-	100%
3	Evaluate data analytics and various supporting services for optimizing real-time IoE decision-making	10%	30%	20%	20%	20%	-	100%
4	Design IOT application for a chosen domain demonstrating integration of sensors, connectivity, and intelligence	10%	20%	20%	20%	10%	20%	100%

IX SUGGESTED LEARNING MATERIALS / TEXTBOOKS / REFERENCE BOOKS

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	IoT Fundamentals: Networking Technologies, Protocols and Use Cases for Internet of Things	1 st Edition	David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Rob Barton and Jerome Henry	Tata Cisco Press	2017
2	Internet of Things – A hands-on approach	1 st Edition	Arshdeep Bahga, Vijay Madiseti	Universities Press	2015
3	Internet of Things: Architecture, Design Principles And Applications	1 st Edition	Rajkamal	McGraw Hill	2017

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Internet of Things for Architects: Architecting IoT solutions by implementing sensors, communication infrastructure, edge computing, analytics, and security,	2nd Edition	Perry Lea	Packt Publications,	2018
2	The Internet of Things: Enabling Technologies, Platforms, and Use Cases	1st Edition	Pethuru Raj, Anupama C. Raman	CRC Press	2017

Online References

Sr. No	Website Name
A	https://owasp.org/www-project-internet-of-things/
B	https://onlinecourses.nptel.ac.in/noc21_ee85/preview , NPTEL Course: Prof. Prabhakar, IISc Bangalore, Design for Internet of Things



Program	Final Year B.Tech. Information Technology (Semester VII)	CL	TL	LL	SL	C
Course Name: Storage Area Network	Course Code: ITDLOCRI7031	3	1	-	2	3
Pre-requisite:	Nil					

I RATIONALE

This subject introduces core concepts of data storage, including NAS, SAN, and storage virtualization. It equips students to manage and secure large-scale digital information effectively. With rising demand for cloud and enterprise storage, these skills are industry-relevant. The course bridges theoretical understanding with practical storage infrastructure applications.

II COMPETENCY

Demonstrate the ability to implement and manage modern storage solutions, including NAS, SAN, and cloud-based virtualization, while ensuring data security, assessing storage needs, and applying appropriate technologies in enterprise environments.

III COURSE OUTCOMES (COs)

After the completion of course based learning Students will be able to,

Course Outcome Number	Course Outcome Statement
1	Evaluate the client-server architecture with emphasis on the necessity of storage-centric and security architectures, including intelligent storage systems and SAN technologies.
2	Evaluate NAS technologies, their role in storage networks, and the I/O technologies integral to SANs.
3	Examine cloud-based storage virtualization methods as defined by SAN standards.
4	Analyze storage infrastructure with respect to security management practices and challenges.

IV TEACHING-LEARNING & ASSESSMENT SCHEME

				Learning Scheme			Assessment Scheme															
Sr. No.	Course Title	Course Type	Course Code	Actual Contact Hrs/Week			SL	Total	Credits	Theory						Practical				SLA		Total
										Exam		CIA		ESE	Total		CIAP	ESEP	Total			
				Duration in Hrs	MSE	CCE		Max		Max	Min	Max	Max		Max	Min	Max	Min				
1	Storage Area Network	DLOC	ITDLOCRI7031	3	1	-	2	90	3	2	20	20	60	100	40	-	-	-	-	25	10	125

V THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
1	CO1	Identify the components and workflow of client-server architecture in distributed systems. Describe the role and features of storage-centric and security architectures in networked environments. Apply basic concepts of intelligent storage systems and SAN technologies in real-world scenarios.	Overview of Networks and Storage Area Networks Integrated storage 1.1 Components of disk drives, RAID levels, storage system environment components, cloud-based software storage provisioning, Systems for Intelligent Storage (ISS), Types of Intelligent Storage Systems. 1.2 Evolution of Storage Systems: Server-Centric IT Architecture's Drawbacks The advantages and disadvantages of SAN-based Storage-Centric IT architecture. Subjects for self-education: Examine This case study on replacing servers with storage networks. 1.3 File Systems: Databases, File Systems, Journaling, Volume Manager Network file systems, Snapshots, and file servers: Performance constraints on file servers, Network Attached Storage (NAS), acceleration of the network file system.	10
2	CO2	List the key components and functions of NAS and SAN architectures. Describe the various I/O technologies used in SAN environments. Demonstrate the use of NAS in different storage network scenarios.	Storage I/O Techniques And Applications 2.1 Fibre Channel Protocol Stack, Physical I/O Path from CPU to Storage System, SCSI, IP storage Infiniband-based storage networks. 2.2 Fibre Channel over Ethernet (FCoE) are some examples of IP storage technologies. applying the NAS Solution to departmental, enterprise, and specialized online workloads, describing NAS workloads; Considerations while integrating SN with NAS Parallels and differences, the need for integration, and the impending integration and connectivity of storage.	11

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Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
3	CO3	Identify different types of storage virtualization techniques used in SANs. Analyze the concept of cloud-based storage and its integration with SAN standards. Illustrate the application of virtualization methods in a cloud storage environment.	Backup and Data Archive and Virtualization 3.1 Overview of Business Continuity: Information Accessibility, Terminology, Planning Life cycle, Business Impact Analysis, and Failure Analysis 3.2 Backup and archiving: objectives, granularity, recovery, techniques, architecture, backup and restore procedures and typologies 3.3 Cloud computing and virtualization: Storage virtualization, SAN virtualization, cloud platforms, and cloud infrastructure virtualization. 3.4 Appliances for virtualization: In-Band, Out-of-Band, and Black Box virtualization Appliances with High Availability for Virtualization and Mass Consumption.	12
4	CO4	List common security threats and vulnerabilities in storage infrastructure. Describe key security management strategies used in protecting storage systems. Apply basic security principles to evaluate the protection level of a given storage setup.	Storage Area Network As a Cloud Service 4.1 Applications that are aware of virtualization, application-aware storage virtualization, and policy-based storage Management is an example of storage automation and virtualization. 4.2 Storage infrastructure security covers a variety of issues, including Securing Storage Infrastructure in Visualized and Cloud Environments 4.3 The tasks involved in managing storage infrastructure and the obstacles encountered in this regard Storage Security Domains, Information Security Framework, and Risk Triad Implementations of Security in Storage Networking Information Life cycle Management, Storage Tearing, and Infrastructure Management for Storage.	12

VI SUGGESTED SELF LEARNING ACTIVITIES

Case Study: Design and Deployment of Secure Storage Infrastructure for real world Scenario.

VII SUGGESTED WEIGHTAGE TO ASSESSMENT PURPOSE

CO No.	Course Outcome	BL1	BL2	BL3	BL4	BL5	BL6	Total
1	Identify the structure and functioning of client-server architecture	10%	30%	60%	-	-	-	100%
2	Evaluate NAS technologies and related I/O in SANs	-	20%	30%	20%	30%	-	100%
3	Examine cloud-based storage virtualization methods	-	20%	30%	30%	20%	-	100%
4	Analyze storage infrastructure security management	-	10%	20%	50%	20%	-	100%

VIII SUGGESTED LEARNING MATERIALS / TEXTBOOKS / REFERENCE BOOKS

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Information Storage and Management	2 nd Edition	G. Somasundaram, Alok Shrivastava, EMC Educational Services	Wiley India	2018
2	Introduction to Storage Area Networks	1 st Edition	Jon Tate, Pall Beck, Hector Hugo Ibarra, Shanmuganathan Kumaravel, Libor Miklas	IBM Redbooks	2017

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Storage Area Network Essentials: A Complete Guide to Understanding and Implementing SANs	3 rd Edition	Richard Barker, Paul Massiglia	Wiley	2010

Online References

Sr. No	Website Name
A	https://education.dell.com/
B	https://nptel.ac.in/courses/106/105/106105182/



Program	Fourth Year B.Tech. Information Technology (Semester VII)	CL	TL	LL	SL	C
Course Name: High Performance Computing	Course Code: ITDLOCRI7032	3	1	-	2	3
Pre-requisite:	Computer Organization, C Programming, Data Structures					

I RATIONALE

This HPC curriculum equips learners with practical and analytical skills needed for modern parallel computing environments. It emphasizes hands-on experience with industry-standard tools and architectures, fostering problem solving, performance tuning, and real-world application development. The modular approach ensures a progressive learning curve from the core principles to emerging technologies.

II COMPETENCY

Develops core competencies in designing, programming, and analyzing parallel computing systems using industry-standard tools like MPI, OpenMP, and CUDA. It fosters the ability to optimize performance across diverse hardware platforms including GPUs and cloud clusters. Learners build problem-solving skills by applying parallel models to real-world applications in AI, scientific computing, and data analytics. The program also nurtures self-directed learning and adaptability in a rapidly evolving HPC landscape.

III COURSE OUTCOMES (COS)

After the completion of course based learning Students will be able to,

Course Outcome Number	Course Outcome Statement
1	Apply key models of parallel computation (SIMD, MIMD, etc.) to categorize existing HPC platforms
2	Apply parallel decomposition techniques to design scalable algorithms.
3	Apply OpenMP and MPI constructs to develop hybrid parallel programs.
4	Apply GPU-based programming techniques (CUDA/OpenCL) to accelerate data-intensive tasks.

IV TEACHING-LEARNING & ASSESSMENT SCHEME

				Learning Scheme						Assessment Scheme													
Sr. No.	Course Title	Course Type	Course Code	Actual Contact Hrs/Week			SL	Total		Credits	Theory					Practical				SLA		Total	
				CL	TL	LL					Exam Duration in Hrs	CIA		ESE	Total			CIAP	ESEP				Total
												Max	CCE		Max	Max	Min			Max	Max		Max
1	High Performance Computing	-	ITDLOCRI7032	3	1	-	2	90	3	2	20	20	60	100	40	-	-	-	-	25	10	125	

V THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
1	CO1	- Differentiate among fundamental parallel computing models and explain their operational characteristics. - Analyze HPC system architectures and classify them based on suitable parallel models. - Apply Flynn's Taxonomy to identify and interpret architectural designs of modern supercomputers and HPC platforms. - Demonstrate model-to-platform mapping by evaluating real-world HPC case studies through the lens of parallel computation frameworks.	Foundations of Parallel & High Performance Computing - Motivation for Parallelism and Scope of HPC - Levels of Parallelism: Instruction, Task, Thread, Memory, Function - Parallel Models: SIMD, MIMD, SIMT, SPMD, Dataflow, Demand-Driven - HPC Architectures: Shared vs Distributed memory, Clusters, Multicores - Communication Overheads and Physical Organization of Platforms	11
2	CO2	- Identify and apply decomposition strategies to divide computational problems into parallel tasks for efficient execution. - Design scalable parallel algorithms by incorporating effective task mapping, load balancing, and communication operations such as broadcast, reduction, and scatter. - Evaluate algorithm performance using metrics like execution time, speedup, efficiency, and cost to guide optimization decisions.	Designing Parallel Algorithms & Measuring Performance - Principles of Parallel Algorithm Design: Decomposition, Task Mapping, Load Balancing - Basic Communication Operations: Broadcast, Reduction, Scatter - Performance Metrics: Execution Time, Speedup, Efficiency, Cost - Scalability Concepts: Granularity, Amdahl's Law, Gustafson's Law - Introduction to Profiling and Performance Bottlenecks	11

(Continued on next page)

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
3	CO3	1. Identify and resolve concurrency issues such as deadlocks and race conditions to ensure the correctness and efficiency of hybrid parallel applications. 2. Design and analyze hybrid programming models (MPI+OpenMP) that exploit both inter-node and intra-node parallelism to enhance scalability across modern HPC architectures.	Parallel Programming Paradigms & Libraries 3.1 Message Passing Interface (MPI): Point-to-Point, Collective Communication 3.2 Topologies and Embedding in MPI 3.3 OpenMP Basics: Directives, Work-Sharing Constructs 3.4 Hybrid Models (MPI+OpenMP): Motivation and Use-cases 3.5 Synchronization, Deadlocks, Race Conditions	11
4	CO4	1. Compare GPU programming models (CUDA vs OpenCL) in the context of device architecture, memory access, and kernel execution strategies. 2. Analyze algorithm performance for applications such as matrix multiplication, graph traversal, and sorting when executed on GPU-based accelerators.	Heterogeneous Computing & Real-world Applications 4.1 GPU Architecture & CUDA Fundamentals: Threads, Warps, Memory 4.2 Open CL Programming Model: Platform, Devices, Kernels 4.3 Parallel Algorithm Case Studies: Matrix Operations, Sorting, Graph Traversals 4.4 HPC in Emerging Fields: Deep Learning, Climate Modeling, Genomics 4.5 Intro to Cloud HPC Platforms (AWS Parallel Cluster, Azure Batch)	12

VI SUGGESTED SELF LEARNING ACTIVITIES

Develop a parallel or heterogeneous computing solution that solves a real-world data processing, simulation, or analytics problem by applying parallel architectures, algorithm design, programming models (MPI/OpenMP/CUDA/OpenCL), and performance evaluation strategies. The solution should demonstrate decomposition, load balancing, communication, and scaling across CPU/GPU/cloud environments.

VII SUGGESTED WEIGHTAGE TO ASSESSMENT PURPOSE

CO No.	Course Outcome	BL1	BL2	BL3	BL4	BL5	BL6	Total
1	Apply key models of parallel computation (SIMD, MIMD, etc.) to categorize existing HPC platforms	-	50%	50%	-	-	-	100%
2	Apply parallel decomposition techniques to design scalable algorithms.	-	30%	40%	30%	-	-	100%
3	Apply OpenMP and MPI constructs to develop hybrid parallel programs.	-	50%	50%	-	-	-	100%
4	Apply GPU-based programming techniques (CUDA/OpenCL) to accelerate data-intensive tasks.	-	30%	40%	30%	-	-	100%

VIII SUGGESTED LEARNING MATERIALS / TEXTBOOKS / REFERENCE BOOKS

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Parallel and High Performance Computing	1 st Edition	R. Robey and Y. Zamora	Manning Publications	2021
2	Introduction to Parallel Computing	2 nd Edition	A. Grama, A. Gupta, G. Karypis, and V. Kumar	Addison-Wesley	2003
3	CUDA Programming: A Developer's Guide to Parallel Computing with GPUs	1 st Edition	S. Cook	Morgan Kaufmann	2013
4	Using MPI: Portable Parallel Programming with the Message-Passing Interface	3 rd Edition	W. Gropp, E. Lusk, and A. Skjellum	MIT Press	2014

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Parallel and High-Performance Computing in Artificial Intelligence	1 st Edition	M. Raghuwanshi, P. Borkar, R. H. Jhaveri, and R. Raut, Eds	Auerbach Publications	2025
2	Programming Your GPU with OpenMP: Performance Portability for GPUs	1 st Edition	T. Deakin and T. Mattson	MIT Press	2023
3	Designing Parallel Algorithms: A Hands-On Approach	1 st Edition	D. Bader	Addison-Wesley	2021
4	Patterns for Parallel Programming	1 st Edition	T. Mattson, B. Sanders, and B. Massingill	Addison-Wesley	2004

**Online References**

Sr. No	Website Name
A	https://onlinecourses.nptel.ac.in/noc25_ma27/preview , NPTEL Course: High Performance Scientific Computing By Prof. Shivasubramanian Gopalakrishnan, Prof. Om Jadhav, Prof. S Vamshi Krishna, Prof. Ashish Kuvelkar (IIT Bombay);
A	https://www.coursera.org/learn/introduction-high-performance-computing , coursera Course: Introduction to High-Performance and Parallel Computing By Shelley Knuth University of Colorado Boulder, Thomas Hauser University of Colorado Boulder;





Program	Final Year B.Tech. Information Technology (Semester VII)	CL	TL	LL	SL	C
Course Name: Infrastructure Security	Course Code: ITDLOCRI7033	3	1	-	2	3
Pre-requisite:	Computer Network, Cryptography & Network Security					

I RATIONALE

Infrastructure Security is crucial for protecting digital assets in today's interconnected world. Protection against a wide range of cyber threats, It also ensures that software is designed and developed with security in mind, addressing vulnerabilities specific to websites and web applications and challenges of cloud computing environments. Wireless security protects wireless networks and devices from unauthorized access and attacks. These security domains are essential for maintaining trust, preventing financial losses, and ensuring business continuity in today's digital landscape.

II COMPETENCY

Analyze cybersecurity fundamentals, including cyberattacks, software security, and core security principles. Skilled in implementing security controls, managing risk, and executing effective incident response. Additionally, demonstrates expertise in cloud security practices and wireless network security protocols..

III COURSE OUTCOMES (COs)

After the completion of course based learning Students will be able to,

Course Outcome Number	Course Outcome Statement
1	Identify software vulnerabilities, cyber attacks, Protection mechanisms.
2	Apply knowledge of security protocols in the context of wireless communication and Cloud infrastructure.
3	Evaluate different attacks on Open Web Applications and Web service.
4	Apply appropriate security policies for protecting infrastructure components.

IV TEACHING-LEARNING & ASSESSMENT SCHEME

				Learning Scheme			Credits	Assessment Scheme														
Sr. No.	Course Title	Course Type	Course Code	Actual Contact Hrs/Week				SL	Total	Exam	Theory					Practical			SLA		Total	
				CL	TL	LL					Duration in Hrs	CIA		ESE	Total	CIAP	ESEP	Total				
								Max	Max			Max	Max					Max	Max	Max		Min
1	Infrastructure Security	-	ITDLOCRI7033	3	1	-	2	90	3	2	20	20	60	100	40	-	-	-	-	25	10	125

V THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
1	CO1	1. Discuss authentication methods. 2. Identify software vulnerabilities. 3. Describe operating system security. 4. Summarize database security.	Introduction 1.1 Cyber-attacks, Vulnerabilities, Defense Strategies and Techniques, Authentication Methods-Password, Token and Biometric, Access Control Policies and Models (DAC,MAC, RBAC, ABAC, BIBA, Bell La Padula) 1.2 Software Vulnerabilities : Buffer overflow, Format String, Cross-Site Scripting, SQL Injection, Malware:Viruses, Worms, Trojans, Logic Bomb, Bots, Rootkits. 1.3 Operating System Security: Memory and Address Protection, File Protection Mechanism, User Authentication. 1.4 Database Security : Database Security Requirements, Reliability and Integrity, Sensitive Data, Inference Attacks, Multilevel Database Security.	10
2	CO2	1. Describe wireless security concept. 2. Apply knowledge of security protocols in the context of cloud infrastructure.	Wireless Security and Cloud Security 2.1 Wireless Security : Mobile Device Security- Security Threats, Device Security, IEEE 802.11xWireless LAN Security, VPN Security, Wireless Intrusion Detection System (WIDS) 2.2 Cloud Security : Cloud Security Risks and Countermeasures, Data Protection in Cloud, Cloud Application Security, Cloud Identity and Access Management, Cloud Security as a Service.	14

(Continued on next page)

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
3	CO3	1. Describe web security consideration, session management, cookies, SSL, HTTPS. 2. Describe Session Hijacking and Management, phishing, Pharming attack, DNS attacks. 3. Evaluate different attacks on Open Web Applications and Web service.	Web Security 3.1 Web Security Considerations, User Authentication and Session Management, Cookies, SSL, HTTPS, SSH, Privacy on Web, Web Browser Attacks, Account Harvesting, Web Bugs, Clickjacking, Cross-Site Request Forgery. 3.2 Session Hijacking and Management, Phishing and Pharming Techniques, DNS Attacks, Web Service Security, Secure Electronic Transaction, Email Attacks. 3.3 Web Server Security as per OWASP, Firewall.	14
4	CO4	1. Describe security policies and business continuity plan. 2. Summarize Indian IT act and Indian Cyber Law.	Information Security and Risk Management 4.1 Security Policies, Business Continuity Plan, Risk Analysis, Incident Management, Legal System and Cybercrime, Ethical Issues in Security Management. 4.2 The Indian IT Act, Indian Cyber Law.	7

VI SUGGESTED SELF LEARNING ACTIVITIES

1. Demonstration of tools Network Security & Monitoring.
2. Demonstration of tools on Vulnerability Scanning.
3. Demonstration of tools on Penetration Testing endenumerate

VII SUGGESTED WEIGHTAGE TO ASSESSMENT PURPOSE

CO No.	Course Outcome	BL1	BL2	BL3	BL4	BL5	BL6	Total
1	Identify vulnerabilities, attacks, Protection mechanisms.	20%	30%	50 %	-	-	-	100%
2	Apply knowledge of security protocols in the context of wireless communication and Cloud infrastructure.	20%	30%	50%	-	-	-	100%
3	Evaluate different attacks on Open Web Applications and Web service.	-	50%	-	-	50%	-	100%
4	Apply appropriate security policies for protecting infrastructure components.	-	50%	50%	-	-	-	100%

VIII SUGGESTED LEARNING MATERIALS / TEXTBOOKS / REFERENCE BOOKS

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Computer Security Principles and Practice	4 th Edition	William Stallings,	Pearson Education	2019
2	Security in Computing	5 th Edition	Charles P. Pfleeger	Pearson Education	2011
3	Network Security and Cryptography	Bernard Menezes	5 th Edition	Cengage Learning	2010

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Web Application Hackers Handbook	4 th Edition	Dafydd Stuttard , Marcus Pinto	Wiley	2011
2	CCNA Security, Study Guide	3 rd Edition	Tim Boyle	Wiley	2010



Mahavir Education Trust's

Shah & Anchor Kutchhi Engineering College

An Autonomous Institute Affiliated to University of Mumbai

Online References

Sr. No	Website Name
A	https://www.coursera.org/learn/introduction-to-cybersecurity-essentials
B	https://www.coursera.org/learn/cloud-security-basics



Program	Final Year B.Tech. Information Technology (Semester VII)	CL	TL	LL	SL	C
Course Name: Software Testing and Quality Assurance	Course Code: ITDLOCRI7034	3	1	-	2	3
Pre-requisite:	Basic knowledge of programming, software engineering principles					

I RATIONALE

The foundation of contemporary software development is now software testing and quality assurance. Ensuring performance, security, and dependability is more important than ever as complex, distributed, and high-stakes software systems proliferate. The STQA syllabus aims to give students exposure to industry-relevant tools and practices, as well as fundamental theoretical foundations and the ability to apply them.

II COMPETENCY

In order to assure safe and dependable software delivery, the STQA develops proficiency in the application of software testing methodologies, automation tools, and quality assurance procedures. It prepares students for professional roles in software quality and testing by allowing them to design test cases, implement automated testing in Agile/DevOps environments, analyze quality metrics, and ensure compliance with industry standards.

III COURSE OUTCOMES (COs)

After the completion of course based learning Students will be able to,

Course Outcome Number	Course Outcome Statement
1	Apply fundamental software testing techniques and strategies to identify defects.
2	Analyze various test design approaches and testing frameworks in modern SDLCs.
3	Evaluate automated testing tools, DevOps pipelines, and quality metrics.
4	Design and implement an end-to-end quality assurance strategy for secure and reliable software delivery.

IV TEACHING-LEARNING & ASSESSMENT SCHEME

				Learning Scheme			Credits	Assessment Scheme																
Sr. No.	Course Title	Course Type	Course Code	Actual Contact Hrs/Week				SL	Total	Exam	Theory					Practical				SLA				Total
				CL	TL	LL					Duration in Hrs	CIA		ESE	Total	CIAP	ESEP	Total						
												MSE	CCE					Max	Min	Max	Min	Max	Min	
1	Software Testing and Quality Assurance	DLOC	ITDLOCRI7034	3	1	-	2	90	3	2	20	20	60	100	40	-	-	-	-	25	10	125		

V THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
1	CO1	1. Explain principles, objectives, and types of software testing. 2. Demonstrate test case design using black-box and white-box techniques. 3. Apply levels of testing (unit, integration, system, acceptance) in structured scenarios.	Fundamentals of Software Testing 1.1 Software Testing Fundamentals- Definitions, Errors, Faults, Failures. 1.2 Software Test Life Cycle (STLC). 1.3 Types of Testing - Functional, Non-functional, Regression, Smoke, Sanity. 1.4 Black-box Testing (Equivalence Partitioning, Boundary Value Analysis). 1.5 White-box Testing (Statement, Branch, Path Testing). 1.6 Test Documentation - Test Plan, Test Case, Defect Report.	11
2	CO2	1. Compare static and dynamic testing techniques. 2. Analyze various test strategies (Agile, Risk-based, Exploratory, Model-based). 3. Interpret test coverage and its significance in SDLC.	Test Design, Strategies, and Modern SDLC Integration 2.1 Static and Dynamic Testing. 2.2 Test Strategies - Agile Testing, Shift-Left Testing, Risk-Based Testing. 2.3 Test Metrics - Code Coverage, Defect Density, Test Efficiency. 2.4 Test Metrics - Code Coverage, Defect Density, Test Efficiency. 2.5 Testing in Agile and Devops Environments.	11
3	CO3	1. Evaluate the role of test automation in CI/CD pipelines. 2. Use industry tools like Selenium, JUnit, and Jenkins for automated testing. 3. Interpret defect prediction and software reliability metrics.	Automation, Tools, and Quality Metrics 3.1 Test Automation Fundamentals—Scope, ROI, Limitations. 3.2 Automation Tools - Selenium WebDriver, TestNG, JUnit, Postman (API Testing). 3.3 Continuous Testing in DevOps — Jenkins, GitHub Actions, and Docker. 3.4 Performance Testing - JMeter, LoadRunner. 3.5 Quality Metrics – MTBF, Defect Leakage, Customer-Found Defects. 3.6 Code Quality - SonarQube, Static Code Analysis.	11

(Continued on next page)

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
4	CO4	1. Construct QA strategies aligned with project and product goals. 2. Identify and mitigate security vulnerabilities using secure testing practices. 3. Design quality standards and audits for continuous improvement.	Quality Assurance Practices and Security Testing 4.1 Quality Assurance vs Quality Control. 4.2 QA Processes - Reviews, Walkthroughs, Audits. 4.3 Secure Software Testing - OWASP Top 10, Penetration Testing Basics. 4.4 Software Configuration Management (SCM). 4.5 Software Process Improvement - CMMI, Six Sigma in QA. 4.6 Industry Standards - ISO 9126, IEEE 829.	12

VI SUGGESTED SELF LEARNING ACTIVITIES

Study ISTQB test strategies; evaluate Agile testing practices using a live GitHub project.

Automate a login scenario using Selenium; explore a CI pipeline using Jenkins or GitHub Actions.

Study OWASP vulnerabilities; design a QA checklist and analyze real-world software failure from a QA perspective.

Watch tutorials on black-box and white-box testing; analyze real-world test cases to identify applied techniques.

VII SUGGESTED WEIGHTAGE TO ASSESSMENT PURPOSE

CO No.	Course Outcome	BL1	BL2	BL3	BL4	BL5	BL6	Total
1	Apply fundamental software testing techniques and strategies to identify defects.	10%	25%	65%	-	-	-	100%
2	Analyze various test design approaches and testing frameworks in modern SDLCs.	-	20%	25%	55%	-	-	100%
3	Evaluate automated testing tools, DevOps pipelines, and quality metrics.	-	-	20%	25%	55%	-	100%
4	Design and implement an end-to-end quality assurance strategy for secure and reliable software delivery.	-	10%	-	20%	25%	45%	100%

VIII SUGGESTED LEARNING MATERIALS / TEXTBOOKS / REFERENCE BOOKS

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Software Testing: Principles and Practices	1 st Edition	Srinivasan Desikan, Gopalaswamy Ramesh	Pearson Education India	2007
2	Foundations of Software Testing	5 th Edition	Rex Black, Dorothy Graham, Erik van Veenendaal	Cengage Learning	2024
3	Software Testing and Quality Assurance: Theory and Practice	1 st Edition	Kshirasagar Naik and Priyadarshi Tripathy	Wiley	2010
4	Agile Testing: A Practical Guide for Testers and Agile Teams	1 st Edition	Lisa Crispin and Janet Gregory	Addison-Wesley	2008

**Reference Books**

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Software Quality Assurance: From Theory to Implementation	2 nd Edition	Daniel Galin	Pearson Education	2018
2	Software Testing: Principles and Practices	2 nd Edition	Dr. Naresh Chauhan	Oxford University Press	2018

Online References

Sr. No	Website Name
A	https://onlinecourses.nptel.ac.in/noc25_cs66/preview
B	https://www.edx.org/learn/software-testing
C	https://www.udemy.com/course/automated-software-testing-with-python/?couponCode=ST4MT20725G1

Program	Final Year B.Tech. Information Technology (Semester VII)	CL	TL	LL	SL	C
Course Name: Mobile Adhoc Network	Course Code: ITDLOCRI7041	2	-	-	2	2
Pre-requisite:	Computer Networks, Cryptography and Network Security.					

I RATIONALE

Mobile Ad Hoc Networks (MANETs) are conceived to provide infrastructure-free, self-organizing wireless networking in environments where fixed infrastructure is unavailable or rapidly deployed (e.g., military operations, disaster zones, rural areas)

II COMPETENCY

Demonstrate the protocols and techniques for efficient communication and routing in Mobile Adhoc Networks (MANETs).

III COURSE OUTCOMES (COs)

After the completion of course based learning Students will be able to,

Course Outcome Number	Course Outcome Statement
1	Analyze the architectural and functional differences between ad hoc wireless networks and traditional cellular networks.
2	Apply the advanced concepts of the MAC layer protocols and different routing protocols for mobile Ad hoc networks.
3	Analyze the Transport layer protocols and Security in Mobile Ad hoc network.
4	Apply the fundamental concepts of QoS, Energy Management and wireless sensor network for Mobile Ad hoc network.

IV TEACHING-LEARNING & ASSESSMENT SCHEME

				Learning Scheme					Assessment Scheme														
Sr. No.	Course Title	Course Type	Course Code	Actual Contact Hrs/Week			SL		Total	Credits		Theory						Practical				SLA	Total
				CL	TL	LL						Duration in Hrs	CIA		ESE	Total		CIAP	ESEP	Total			
													Max	CCE		Max	Min			Max	Max		
1	Mobile Adhoc Network	Program Core Course (PC)	ITDLOCRI7041	2	-	-	2	60	2	2	-	50	-	50	20	-	-	-	-	25	10	75	

V THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
1	CO1	1. Describe the fundamentals Cellular and Ad Hoc Wireless Networks. 2. Discuss the Concepts and architectures of Mobile Ad hoc networks (MANETs) and wireless sensor networks (WSN) 3. Discuss the Applications of Ad hoc network and Design Challenges in Ad hoc network.	Introduction to Ad hoc Wireless Networks 1.1 Introduction: Cellular and Ad Hoc Wireless Networks, Fundamentals of Wireless Communication Technology: The Electromagnetic Spectrum, Characteristics of the Wireless Channel. 1.2 Concepts and architectures of Mobile Ad hoc networks (MANETs) 1.3 Applications of Ad hoc network, Design Challenges in Ad hoc network.	4
2	CO2	1. Discuss the Issues in designing a MAC Protocol and Design Goals of MAC Protocols. 2. Describe the Routing Protocols in Ad-hoc Wireless Networks 3. Discuss the Multicast Routing in Ad-hoc Wireless Networks.	Medium Access Control Protocols and Routing protocols 2.1 Issues in Designing a MAC Protocol, Design Goals of MAC Protocols, Classification of MAC protocols, Contention-Based Protocols with Reservation Mechanisms and Scheduling Mechanisms, IEEE 802.11a and HiperLan standard. 2.2 Routing Protocols in Ad-hoc Wireless Networks: Introduction, Design issues, Classification of Routing Protocols: Routing information update mechanism, Routing topology. 2.3 Multicast Routing in Ad-hoc Wireless Networks: Introduction, Design Issues, Operation of Multicast Routing Protocols, An Architecture Reference Model for Multicast Routing Protocols.	10

(Continued on next page)

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
3	CO3	1. Describe the Transport Layer Protocol in Ad-hoc Wireless Networks. 2. Discuss the Issues and Challenges in Security Provisioning in Ad-hoc Wireless Networks.	Transport Layer and Security Protocols 3.1 Transport Layer in Ad-hoc Wireless Networks: Introduction, Design Issues and Goals of a Transport Layer Protocol; Classification of Transport Layer Solutions. 3.2 Security in Ad-hoc Wireless Networks: Issues and Challenges in Security Provisioning, Network Security Attacks classification.	7
4	CO4	1. Discuss Quality of Service in Ad-hoc Wireless Networks. 2. Describe Energy Management in Ad-hoc Wireless Networks 3. Discuss the fundamentals of Wireless Sensor Networks.	Quality of Service, Energy Management and Wireless Sensor Networks 4.1 Quality of Service in Ad-hoc Wireless Networks: Introduction, Issues and Challenges in Providing QoS in Ad-hoc Wireless Networks, Classification of QoS Solutions. 4.2 Energy Management in Ad-hoc Wireless Networks: Introduction, Need for Energy Management in Ad-hoc Wireless Networks, Classification of Energy Management Schemes. 4.3 Wireless Sensor Networks (WSN): Introduction, Sensor Network Architecture, Applications of WSN, Data Dissemination, Data Gathering.	9

VI SUGGESTED SELF LEARNING ACTIVITIES

Create vlog and study report for the different scenario in MANET like Routing Protocol Selection, Mobility Model Design, Security Considerations, Performance Evaluation, Energy Efficiency Strategy etc

VII SUGGESTED WEIGHTAGE TO ASSESSMENT PURPOSE

CO. No.	Course Outcome	BL1	BL2	BL3	BL4	BL5	BL6	Total
1	Analyze the architectural and functional differences between ad hoc wireless networks and traditional cellular networks.	-	10%	40%	50%	-	-	100%
2	Apply the advanced concepts of the MAC layer protocols and different routing protocols for mobile Ad hoc networks.	-	40%	60%	-	-	-	100%
3	Analyze the Transport layer protocols and Security in Mobile Ad hoc network.	-	25%	25%	50%	-	-	100%
4	Apply the fundamental concepts of QoS, Energy Management and wireless sensor network for Mobile Ad hoc network.	-	40%	60%	-	-	-	100%

VIII SUGGESTED LEARNING MATERIALS / TEXTBOOKS / REFERENCE BOOKS

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Ad Hoc Wireless Networks: Architectures and Protocols	2 nd Edition	C. S. Ram Murthy, B. S. Manoj	Prentice Hall of India	2005
2	Adhoc Mobile Wireless Networks	2 nd Edition	C. K. Toh	Pearson Education	2002
3	Wireless Communications and Networks	2 nd Edition	William Stallings	Pearson Education	2004

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Wireless Ad Hoc Networking: Personal-Area, Local-Area, and the Sensory-Area Networks	1 st Edition	Shih-Lin Wu Yu-Chee Tseng	Auerbach Publications	2007
2	Adhoc Mobile Wireless Network: Principles, Protocols and Applications	2 nd Edition	Subir Kumar Sarkar	CRC Press	2007
3	Ad Hoc Networks: Technologies and Protocols	2 nd Edition	Prashant Mohapatra and Sriramamurthy	Springer International Edition	2004

Online References

Sr. No	Website Name
A	https://onlinecourses.nptel.ac.in/noc25_cs74/preview , NPTEL Course: Wireless Ad Hoc and Sensor Networks By Prof. Sudip Misra (IIT Kharagpur)
B	https://www.udemy.com/course/wireless-and-mobile-communication-advanced-level-i/?srsltid=AfmB0ooaOPrdEwTU3PMFpl5oggnQsukyxxvmrQDWQQhGDwN2K2QMUey1v , Udemy Course: Wireless and Mobile Communication- Advanced Level-I By Dr Usha G
C	https://nptel.ac.in/courses/106105160 , NPTEL Course: Wireless Ad Hoc and Sensor Networks, Prof. Sudip Misra (IIT Kharagpur)



Program	Final Year B.Tech. Information Technology (Semester VII)	CL	TL	LL	SL	C
Course Name: Augmented & Virtual Reality	Course Code: ITDLOCRI7042	2	-	-	2	2
Pre-requisite:	Computer Graphics					

I RATIONALE

Integrating Augmented Reality (AR) and Virtual Reality (VR) into modern systems and experiences lies in their transformative potential to reshape how we perceive, interact with, and learn from digital content.

II COMPETENCY

Develop immersive AR and VR technologies to enhance user engagement, enable interactive experiences, and drive innovation across industries such as education, healthcare, entertainment, retail, and engineering.

III COURSE OUTCOMES (COs)

After the completion of course based learning Students will be able to,

Course Outcome Number	Course Outcome Statement
1	Identify, examine, and develop software that reflects fundamental techniques for the design and deployment of VR and AR experiences
2	Describe how VR and AR systems work
3	Choose, develop, defend the use of particular designs for AR and VR experiences
4	Examine AR and VR design problems and solutions from the industry and academic

IV TEACHING-LEARNING & ASSESSMENT SCHEME

				Learning Scheme			Credits	Assessment Scheme																			
Sr. No.	Course Title	Course Type	Course Code	Actual Contact Hrs/Week				SL	Total	Theory	Practical						SLA		Total								
				CL	TL	LL					CIAP	ESEP	Total														
													Duration in Hrs	MSE	CCE	ESE				Total		Max	Min	Max	Min	Max	Min
																				Max	Min						
1	Augmented & Virtual Reality	-	ITDLOCRI7042	2	-	-	2	60	2	2	-	50	-	50	20	-	-	-	-	25	10	75					

V THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
1	CO1	1. Define and distinguish between Augmented Reality (AR), Virtual Reality (VR), and Mixed Reality (MR). 2. Explain the taxonomy of immersive technologies and identify where AR, VR, and MR fall on the virtuality continuum.	Introduction of Augmented-Virtual and Mixed Reality 1.1 An Overview of Mixed Reality and Augmented Virtual Reality Taxonomy, features, and technologies of augmented reality; distinctions among AR, VR, and MR; 1.2 Issues with augmented reality, its systems and functioning 1.3 Methods for augmented reality and approaches for visualizing it.	10
2	CO2	1. Explain the fundamental characteristics of VR systems, including immersion, presence, interactivity, and real-time feedback. 2. Classify various VR output devices, focusing on visual display technologies like HMDs, stereoscopic displays, and retinal projection systems. 3. Describe the role of game engines (e.g. Unity, Unreal Engine) in VR content creation.	VR System and VR Software Development 2.1 VR as an academic field, fundamental characteristics of VR systems, VR system architecture and hardware. 2.2 VR output hardware: visual displays, VR input hardware: tracking systems, motion capture systems, data gloves. 2.3 VR software development challenges Game engines, master/slave and client/server topologies 2.4 Cluster rendering, and SDKs that are available to create VR apps for various hardware	10

(Continued on next page)

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
3	CO3	1. Describe core activities in 3D manipulation, such as selection, translation, rotation, and scaling of virtual objects. 2. Analyze usability trade-offs of techniques like raycasting, virtual hand, and Go-Go techniques in different immersive scenarios. 3. Understand and perform camera calibration, including intrinsic and extrinsic parameter tuning, to improve tracking accuracy and spatial alignment.	3D interaction techniques and AR software development 3.1 Activities involving 3D manipulation, manipulation methods, and input devices. 3.2 3D manipulation interaction strategies. 3.3 Augmented reality software, camera calibration and settings, AR Toolkit, augmented reality based on markers.	5
4	CO4	1. Evaluate the creative and technical benefits of VR in film/TV production, such as enhanced audience immersion, virtual set visualization, and real-time collaboration between crew and directors.	Applications of VR 4.1 VR Technology's Applicability in Digital Entertainment: VR Production for Film and Television. 4.2 VR technology for games and physical exercise. Virtual reality demonstration of digital entertainment.	5

VI SUGGESTED SELF LEARNING ACTIVITIES

- 1.Task: Research and summarize the following topics in your own words: What are 3D interaction techniques? Give at least three real-world use cases (e.g. VR training simulations). Compare AR vs. VR: How do interaction techniques differ?
- 2.Design your own AR application idea (e.g., AR navigation for malls or museums). The proposal should include the following: App concept and user needs Key 3D interactions Technologies/tools you would use Sketches or wireframes

VII SUGGESTED WEIGHTAGE TO ASSESSMENT PURPOSE

Co No.	Course Outcome	BL1	BL2	BL3	BL4	BL5	BL6	Total
1	Identify, examine, and develop software that reflects fundamental techniques for the design and deployment of VR and AR experiences	-	25%	50%	25%	-	-	100%
2	Describe how VR and AR systems work	-	25%	50%	25%	-	-	100%
3	Choose, develop, defend the use of particular designs for AR and VR experiences	-	25%	50%	25%	-	-	100%
4	Examine AR and VR design problems and solutions from the industry and academic	-	25%	50%	25%	-	-	100%

VIII SUGGESTED LEARNING MATERIALS / TEXTBOOKS / REFERENCE BOOKS

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Human-Centered Design for Virtual Reality	3 rd Edition	Jason Jerald	Association for Computing Machinery and Morgan	2015
2	Understanding Augmented Reality, Concepts and Applications	4 th Edition	Tony Parisi	O'Reilly	2015

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Developing Virtual Reality Applications", Foundations of Effective Design,	1 st Edition	Alan B. Craig, Jeffrey D. Will, and Sherman William	Elsevier	2009

Online References

Sr. No	Website Name
A	https://onlinecourses.nptel.ac.in/noc25_cs87/preview , NPTEL Course: Foundations of Virtual Reality, By Prof. M.Manivannan, Prof. Anna Lavalle, Prof. Steven Lavalle (IIT Madras);



Program	Final year B.Tech. Information Technology (Semester VII)	CL	TL	LL	SL	C
Course Name: Quantum Computing	Course Code: ITDLOCRI7043	2	-	-	2	2
Pre-requisite:	NIL					

I RATIONALE

Quantum computers have the potential to solve certain classes of problems exponentially faster than classical computers. For instance, Shor's algorithm can factor large integers exponentially faster than the best-known classical algorithms, posing both a challenge and opportunity for cyber security. Similarly, Grover's algorithm provides quadratic speedup for unstructured search problems.

II COMPETENCY

Demonstrate understanding and application of the fundamental principles of quantum mechanics to develop and analyze quantum algorithms, simulate quantum systems, and evaluate the potential and limitations of quantum computing in solving real-world problems.

III COURSE OUTCOMES (COs)

After the completion of course based learning Students will be able to,

Course Outcome Number	Course Outcome Statement
1	Apply quantum algorithms to solve computational problems across different domains.
2	Illustrate the impact of quantum noise on quantum systems and interpret quantum operations.
3	Apply suitable error correction techniques to simple quantum circuits.
4	Apply suitable quantum cryptographic algorithm .

IV TEACHING-LEARNING & ASSESSMENT SCHEME

				Learning Scheme			Credits	Assessment Scheme														
Sr. No.	Course Title	Course Type	Course Code	Actual Contact Hrs/Week				SL	Total		Theory					Practical				SLA		Total
				CL	TL	LL				Exam	CIA		ESE	Total			CIAP	ESEP	Total			
											Duration in Hrs	MSE		CCE	Max	Min			Max	Min	Max	
1	Quantum Computing	-	ITDLOCRI7043	2	-	-	2	60	2	-	-	50	-	50	20	-	-	-	-	25	10	75

V THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
1	CO1	1. Summarize fundamentals of quantum computing. 2. Apply quantum operations such as single qubit gates, control gates, and measurement techniques to solve problems involving quantum circuits. 3. Describe the working of quantum search algorithms and their advantages in solving NP-complete problems, unstructured database searches, and quantum counting. 4. Construct quantum circuits using Boolean functions and universal quantum gates for basic algorithm implementation.	Fundamentals of Quantum Computing 1.1 Fundamental Concepts: Introduction and Overview – Global Perspectives – Quantum Bits – Quantum Computation – Quantum Algorithms – Experimental Quantum Information Processing – Quantum Information. 1.2 Problems on Qubits, Quantum Circuits – Quantum algorithms, Single Orbit operations, Control Operations, Measurement, Universal Quantum Gates, Simulation of Quantum Systems, Quantum Fourier transform, Phase estimation. 1.3 Applications, Quantum search algorithms – Quantum counting – Speeding up the solution of NP – complete problems – Quantum Search for an unstructured database. 1.4 Problems on Boolean functions and Quantum gates, Quantum gates and circuits.	08
2	CO2	1. Describe the guiding principles and physical implementations of quantum computation, including harmonic oscillator-based, optical photon-based, and ion trap systems. 2. Apply quantum operation models to analyze the effects of noise on quantum states using representative examples. 3. Solve problems related to quantum measurements and quantify the impact using appropriate distance metrics for quantum information.	Quantum Computers and Information 2.1 Guiding Principles, Conditions for Quantum Computation, Harmonic Oscillator Quantum Computer, Optical Photon Quantum Computer – Optical cavity Quantum electrodynamics, Ion traps, Nuclear Magnetic resonance. 2.2 Quantum noise and Quantum Operations – Classical Noise and Markov Processes, Quantum Operations, Examples of Quantum noise and Quantum Operations. 2.3 Applications of Quantum operations, Limitations of the Quantum operations formalism, Distance Measures for Quantum information. Problems on Measurement.	08

(Continued on next page)

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
3	CO3	- Summarize the fundamentals of quantum error correction including Shor code, stabilizer codes, and the principles behind constructing quantum codes. - Apply quantum error correction techniques to protect quantum information in example scenarios.	Quantum Error Correction 3.1 Introduction, Shor code, Theory of Quantum Error –Correction, Constructing Quantum Codes, Stabilizer codes. 3.2 Fault – Tolerant Quantum Computation, Case study on Quantum error correction.	07
4	CO4	- Apply basic cryptographic methods to secure quantum communication using privacy amplification and reconciliation techniques. - Summarize the concepts of quantum key distribution. - Solve problems involving quantum error correction techniques and cryptographic protocols to protect quantum data.	Quantum Cryptography 4.1 Quantum Cryptography-Private Key Cryptography, Privacy Amplification and Information Reconciliation. 4.2 Quantum Key Distribution, Privacy and Coherent Information, The Security of Quantum Key Distribution. 4.3 Problems on Quantum error correction and cryptography.	07

VI SUGGESTED SELF LEARNING ACTIVITIES

Conduct detailed study based on Quantum Computing for real time applications

Suggested list

- Quantum Key Distribution (QKD) for Secure Communications
- China's Quantum Satellite "Micius"
- Hybrid classical-quantum algorithms (e.g., QAOA, VQE)
- Quantum machine learning (QML) models and frameworks

VII SUGGESTED WEIGHTAGE TO ASSESSMENT PURPOSE

CO No.	Course Outcome	BL1	BL2	BL3	BL4	BL5	BL6	Total
1	Apply quantum algorithms to solve computational problems across different domains.	-	40%	60%	-	-	-	100%
2	Illustrate the impact of quantum noise on quantum systems and interpret quantum operations.	-	50%	50%	-	-	-	100%
3	Apply suitable error correction techniques to simple quantum circuits.	-	50%	50%	-	-	-	100%
4	Apply suitable quantum cryptographic algorithm .	-	50%	50%	-	-	-	100%

VIII SUGGESTED LEARNING MATERIALS / TEXTBOOKS / REFERENCE BOOKS

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Introduction to Quantum Computing	2 nd	Ray LaPierre	Springer / MRS Series	June 2025
2	Quantum Computation and Quantum Information	10 th	M.A. Nielsen and I.Chuang	Cambridge University Press	2010
3	Problems and Solutions in Quantum Computing and Quantum Information	4 th	Willi-Hans Steeb and Yorick Hardy	World Scientific (distributed by Dev Publishers in South Asia)	2020
4	Quantum Computing for Everyone	4 th	Chris Bernhardt	MIT Press	September 2020.

**Reference Books**

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Quantum Computing: A Beginner's Introduction	1 st	Parag K. Lala	Tata McGraw Hill	2020
2	Quantum Computing	-	V. Sahni	Tata McGraw Hill	2007

Online References

Sr. No	Website Name
A	https://nptel.ac.in/courses/106106232 , NPTEL Course: Introduction to Quantum Computing: Quantum Algorithms and Qiskit, IBM and IITM



Program	Final Year B.Tech. Information Technology (Semester VII)	CL	TL	LL	SL	C
Course Name: Information Retrieval System	Course Code: ITDLOCRI7044	2	-	-	2	2
Pre-requisite:	Data Structure and Analysis of Algorithm					

I RATIONALE

As there is exponential growth of data in various domains like web, academic research, healthcare, finance, and social media there is a need for efficient systems to search, retrieve, and rank relevant information. Information Retrieval (IR) Systems provide students with a comprehensive understanding of the principles, models, and technologies behind modern IR systems. Topics such as indexing, ranking algorithms, relevance feedback and evaluation metrics which equip learners with theoretical foundation.

II COMPETENCY

Apply foundational models, indexing strategies, and query processing techniques to design, build, and assess information retrieval systems capable of extracting relevant data from large-scale, unstructured information sources using standard evaluation metrics.

III COURSE OUTCOMES (COs)

After the completion of course based learning Students will be able to,

Course Outcome Number	Course Outcome Statement
1	Apply the basic concepts of Information Retrieval for document indexing and query processing.
2	Solve text and multimedia retrieval queries along with their associated operations.
3	Demonstrate various indexing and searching techniques.
4	Design the user interface for an information retrieval system.

IV TEACHING-LEARNING & ASSESSMENT SCHEME

				Learning Scheme				Credits	Assessment Scheme													
Sr. No.	Course Title	Course Type	Course Code	Actual Contact Hrs/Week			SL		Total	Exam	Theory					Practical				SLA		Total
				CL	TL	LL					Duration in Hrs	CIA		ESE	Total			CIAP	ESEP			
												Max	Max		Max	Max	Min	Max	Max	Max	Min	
1	Information Retrieval System	DLOC-4	ITDLOCRI7044	2	-	-	2	60	2	2	-	50	-	50	20	-	-	-	-	25	10	75

V THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
1	CO1	- Analyse the process of information retrieval system with its components and types. - Differentiate between information retrieval and data retrieval systems. - Compare the functionalities of search engines and web browsers.	Information Retrieval System Fundamentals 1.1 Motivation, Basic Concepts, The Retrieval Process 1.2 Information System: Components and parts, Types of Information system 1.3 Definition and objectives on information retrieval system. Information versus Data Retrieval. 1.4 Search Engines and browsers, Search API.	7
2	CO2	- Demonstrate various query techniques while utilizing suitable query languages and Query protocols. - Implement user relevance feedback mechanisms in query operations. - Implement multimedia information retrieval models by utilizing text and multimedia languages. - Implement document pre-processing techniques.	Query and Text Processing 2.1 Query Languages, Keyword based Querying, Structural Queries, Query Protocols, Pattern Matching, Proximity, Queries and Wildcard Queries. 2.2 Query Operations: User relevance feedback 2.3 Multimedia IR models: Data Modelling, Text and Multimedia languages and properties, Metadata, Mark-up Languages. 2.4 Text Operations: Document Pre-processing, Document Clustering	8
3	CO3	- Apply efficient text retrieval strategies by utilizing query processing techniques. - Design multimedia indexing systems by applying generic indexing frameworks. - Evaluate the complexities of web search by characterizing the structure and dynamics of the web.	Indexing and Searching 3.1 Inverted files, other indices for text, Boolean Queries, Structural Queries, Text compression techniques. 3.2 Indexing and Searching: - A Generic Multimedia indexing approach, Automatic Feature extraction. 3.3 Searching Web: Challenges, Characterizing the web, Browsing, Searching using Hyperlinks, Meta searches.	7

(Continued on next page)

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
4	CO4	- Summarize the significance of effective User Interface design. - Analyze the role of human-computer interaction (HCI) in the design process. - Analyze the stages of the information access process in human-computer interaction. - Assess user interfaces support for information search process.	User interface and Visualization 4.1 Importance of user Interface – definition, importance of good design. Benefits of good design. A brief history of Screen design. 4.2 The graphical user interface – popularity of graphics, the concept of direct manipulation, graphical system. 4.3 Design process – Human interaction with computers, importance of human characteristics, human Consideration, Human interaction speeds, and understanding business junctions. 4.4 Human Computer interaction, the information access process, starting points, query specifications. 4.5 Interface support for the search process, using relevance judgments, context.	8

VI SUGGESTED SELF LEARNING ACTIVITIES

- Detail study on collection of results from multiple search engine and analyze how meta-search improves coverage.
- Explore and implement effective text processing techniques to clean, normalize, and prepare Twitter text data for downstream analysis tasks such as sentiment analysis, topic modelling, and trend detection.
- Micro project: To design and implement a basic information retrieval system that indexes and searches a collection of documents from a specific domain such as movie reviews, academic abstracts, or news articles.

VII SUGGESTED WEIGHTAGE TO ASSESSMENT PURPOSE

Sr. No.	Course Outcome	BL1	BL2	BL3	BL4	BL5	BL6	Total
1	Apply the basic concepts of Information Retrieval for document indexing and query processing.	20%	30%	50%	-	-	-	100%
2	Solve text and multimedia retrieval queries along with their associated operations.	20%	30%	50%	-	-	-	100%
3	Demonstrate various indexing and searching techniques.	-	20%	30%	50%	-	-	100%
4	Design the user interface for an information retrieval system.	20%	30%	50%	-	-	-	100%

VIII SUGGESTED LEARNING MATERIALS / TEXTBOOKS / REFERENCE BOOKS

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Modern Information Retrieval	2 nd Edition	Berthier Ribeiro- Neto	ACM Press-Addison Wesley	2011
2	Information Retrieval Systems: Theory and Implementation	2 nd Edition	Gerald Kowaski	Kluwer Academic Publisher	2000
3	Storage Network Management and Retrieval	2 nd Edition	Dr. Vaishali Khairnar, Nilima Dongre	Wiley India.	2015
4	Human – Computer Interaction	3 rd Edition	Alan Dix, Janet Fincay, Gregory's, Abowd, Russell Bealg,	Pearson Education	2003

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Introduction to Information Retrieval	1 st Edition	Christopher D. Manning and Prabhakar Raghavan	Cambridge University Press	2008
2	Introduction to Modern Information Retrieval	3 rd Edition	G.G. Chowdhury	Neal-Schuman Publishers	2010

Online References

Sr. No	Website Name
A	https://www.coursera.org/learn/text-retrieval
B	https://www.udemy.com/course/information-retrieval-and-mining-massive-data-sets



Program	Final Year B.Tech (Semester VII)	CL	TL	LL	SL	C
Course Name: Management Information System	Course Code: ECILOCR1705 (ILOC1)	3	1	-	2	3
Course Type:	-					
Pre-requisite:	Database Management System ,Computer Network, Software Engineering					

I RATIONALE

The Management Information System (MIS) course helps students to bridge the gap between technology and managerial decision-making in modern, data-driven businesses. It focuses on how information systems support key business functions and strategic goals through tools like ERP, CRM, and analytics. Students explore emerging technologies such as Cloud Computing, IoT, and while also understanding cyber security and ethical data use.

II COMPETENCY

Manage, analyze, and apply information systems to support decision-making, improve business processes, and gain strategic advantage in an organization.

III COURSE OUTCOMES (COs)

After the completion of course based learning Students will be able to,

Course Outcome Number	Course Outcome Statement
1	Analyze business analytics tools and techniques to convert data into actionable insights for effective decision-making.
2	Examine ethical issues and security threats to the organization to protect information resources.
3	Analyze customer behavior and segmentation strategies to improve customer Engagement.
4	Select relevant IT solutions to the business over traditional methodologies.

IV TEACHING-LEARNING & ASSESSMENT SCHEME

				Learning Scheme			Assessment Scheme															
Sr. No.	Course Title	Course Type	Course Code	Actual Contact Hrs/Week			SL	Total	Credits	Exam	Theory					Practical				SLA		Total
				CL	TL	LL					Duration in Hrs	CIA		ESE	Total	CIAP	ESEP	Total				
												MSE	CCE									
							Max	Max				Max	Max	Min	Max	Max	Max	Min	Max	Min		
1	Management Information System	-	ECILOCR1705	3	1	-	2	90	3	2	20	20	60	100	40	-	-	-	-	25	10	125

V THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
1	CO1	1. Define the basic concepts of Information Systems (IS) and Computer-Based Information Systems. 2. Differentiate between the traditional file approach and database approach. 3. Apply Business Intelligence techniques to analyze business data. 4. Analyze data patterns and trends for informed decision-making	Introduction To Information Systems (IS), Data and Knowledge Management and Business Analytics 1.1 Information Systems (IS), Computer Based Information Systems, Impact of IT on organizations, Importance of IS to Society. Organizational Strategy, Competitive Advantages and Strategic IS. 1.2 Data and Knowledge Management: Managing Data, Database Approach, Big Data, Data warehouse and Data Marts, Knowledge Management 1.3 Business Analytics: Managers and Decision Making, Business Intelligence (BI), BI for Data analysis.	11
2	CO2	1. Define ethical principles and their relevance to information systems. 2. Identify ethical dilemmas and issues related to the use of IT and data. 3. Evaluate real-world scenarios involving ethical violations in IT usage. 4. Apply basic security controls and best practices to secure digital information. 5. Analyze organizational vulnerabilities and recommend appropriate security measures.	Ethics and Privacy, Information Security 2.1 Ethical Issues, Privacy 2.2 Information Security, Unintentional and Deliberate Threats to Information Systems, Protection about Information Resources, Information Security Controls	9

(Continued on next page)

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
3	CO3	- Describe wired and wireless network technologies used in business environments. - Analyze the benefits, concerns, and risks associated with adopting Cloud Computing. - Analyze how E-Business strategies transform business operations. - Compare different CRM and SCM tools and their features. - Apply SC tools for improving customer interaction and employee engagement.	Computer Networks, E-Business and E-Commerce and Social Computing 3.1 Computer Networks Wired and Wireless technology, The Internet of Things, Wireless Security, Cloud computing (features, benefits, types, services, concerns and risks with CC) 3.2 E-business and E-commerce – B2B, B2C. Mobile commerce. 3.3 Customer Relationship Management and Supply Chain Management. 3.4 Social Computing (SC), Web 2.0 and 3.0, SC in Business: Shopping, Marketing, CRM, HRM	15
4	CO4	- Explain the role of TPS and FAIS in supporting business operations. - Analyze the benefits and challenges of implementing ERP in organizations. - Apply cost-benefit analysis techniques for evaluating IT proposals. - Compare various strategies for acquiring IT applications (in-house development, outsourcing, etc.). - Explain the purpose and activities of each SDLC phase. - Compare SDLC with alternative systems development approaches (e.g., Agile, RAD, Prototyping).	Information Systems Within the Organization and Acquiring Information Systems and Applications 4.1 Transaction Processing Systems, Functional Area Information Systems, Enterprise Resource Planning Systems, ERP Support for Business Processes 4.2 Planning for and Justifying IT Applications, Strategies for Acquiring IT Applications, Traditional Systems Development Life Cycle, Alternative Methods and Tools for Systems Development.	10

VI SUGGESTED SELF LEARNING ACTIVITIES

Case study.

VII SUGGESTED WEIGHTAGE TO ASSESSMENT PURPOSE

CO.No	Course Outcome	BL1	BL2	BL3	BL4	BL5	BL6	Total
1	Analyze business analytics tools and techniques to convert data into actionable insights for effective decision-making.	-	20%	40%	40%	-	-	100%
2	Examine ethical issues and security threats to the organization to protect information resources.	-	-	50%	50%	-	-	100%
3	Analyze customer behavior and segmentation strategies to improve customer Engagement.	-	-	60%	40%	-	-	100%
4	Select relevant IT solutions to the business over traditional methodologies.	-	-	60%	40%	-	-	100%

VIII SUGGESTED LEARNING MATERIALS / TEXTBOOKS / REFERENCE BOOKS

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Management Information Systems	4 th Edition	Kelly Rainer, Brad Prince	Wiley Publication	2016
2	Management Information Systems: Managing the Digital Firm	10 th Edition	K.C. Laudon and J.P. Laudon	Prentice Hall	2007
3	Managing Information Systems: Strategy and Organization	3 rd Edition	D. Boddy, A. Boonstra	Prentice Hall	2009



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Online References

Sr. No	Website Name
A	https://onlinecourses.nptel.ac.in/noc20_mg60/preview , NPTEL Course:Management Information System By Prof.Kunal Kanti Ghosh, Prof.Saini Das, Prof.Surojit Mukherjee,IIT Kharagpur



Program	Final Year B.Tech. (Semester VII)	CL	TL	PL	SL	C
Course Name: Research Methodology	Course Code: CSILOC1705	3	1	-	2	3
Course Type:	-					
Pre-requisite:	-					

I RATIONALE

Research Methodology helps students develop analytical thinking, problem-solving skills, and a systematic approach to research. It prepares them for innovation, higher studies, and evidence-based decision-making in both academic and industrial settings.

II COMPETENCY

Analyze research concepts, formulate research problems, design appropriate methodologies, collect and examine data, and interpret findings to produce ethical and meaningful outcomes addressing real-world challenges.

III COURSE OUTCOMES (COs)

After the completion of course-based learning, students will be able to:

Course Outcome Number	Course Outcome Statement
1	Apply the basic concepts, types, and significance of research in business and social sciences.
2	Analyze appropriate research and sampling designs to address business and social science problems.
3	Evaluate various stages of research methodology including literature, hypothesis, and data.
4	Create a structured research report addressing formulation, ethical concerns, and outcomes.

IV TEACHING-LEARNING & ASSESSMENT SCHEME

				Learning Scheme			Credits	Assessment Scheme															
Sr. No.	Course Title	Course Type	Course Code	Actual Contact Hrs/Week				SL	Total	Paper	Theory					Practical				SLA		Total	
											Duration in Hrs	CIA		ESE	Total			CIAP	ESEP				Total
				MSE	CCE	Max		Min	Max			Min	Max		Min								
				Notional Hours				Max	Max			Max	Max		Min	Max	Min						
1	Research Methodology	-	CSILOC1705	3	1	-	2	90	3	2	20	20	20	60	100	40	-	-	-	-	25	10	125

V THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Module	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
1	CO1	1. Apply basic research terms like construct, postulate, and hypothesis. 2. Analyze the distinction between research methods and methodology. 3. Evaluate the need and objectives of research in different fields. 4. Create classifications of research types and approaches.	Fundamental research concepts, methods, objectives, and various research types and approaches. 1.1 Research Concepts and Definitions 1.2 Research Methods vs Methodology 1.3 Need and Objectives of Research 1.4 Types of Research and Approaches	15
2	CO2	1. Apply knowledge of research design to structure investigations. 2. Evaluate essentials of effective sampling in research. 3. Analyze different sampling techniques and their applications. 4. Create a sample design considering errors and stages involved.	Research Design and Sample Design 2.1 Research Design 2.2 Essentials of Sampling 2.3 Sampling Techniques 2.4 Sample Design Process	15
3	CO3	1. Apply the concept of research methodology in structuring academic studies. 2. Create a logical sequence for the scientific research process. 3. Analyze and select relevant research problems. 4. Evaluate literature to formulate a valid hypothesis.	Research Methodology 3.1 Meaning of Research Methodology 3.2 Scientific Research Process 3.3 Research Problem Identification 3.4 Literature Review and Hypothesis Formulation	15
4	CO4	1. Evaluate the feasibility and relevance of a research problem. 2. Apply data selection and analysis strategies effectively. 3. Create a report based on valid conclusions and interpretations. 4. Analyze ethical issues and provide research-based recommendations.	Formulating Research Problem and Outcome of Research 4.1 Formulating Research Problem 4.2 Data Analysis and Interpretation 4.3 Report Writing 4.4 Ethics and Recommendations	15

VI SUGGESTED SELF LEARNING ACTIVITY

- Obtain a professional certification by completing a relevant technical course from a recognized platform (e.g., Cisco, Infosys Springboard, NPTEL/SWAYAM, Coursera, edX, Udemy). Demonstrate completion by submitting the official certificate and a concise report on the key skills and concepts acquired.

OR

- Develop an engaging video tutorial on a fundamental subject topic, suitable for publication on the department's official social media platforms.

OR

- Research and present a real-world case study that is directly relevant to the subject, highlighting how subject concepts apply to the scenario.

OR

- Participate in any subject-related event (hackathon/ideathon/codeathon/technical meetup/conference, etc) and submit valid proof of participation.

OR

- Review five recent papers from Google Scholar in your domain.

VII SUGGESTED WEIGHTAGE TO ASSESSMENT PURPOSE

CO. No.	Course Outcome	BL1	BL2	BL3	BL4	BL5	BL6	Total
1	Apply the basic concepts, types, and significance of research in business and social sciences.	-	10%	40%	30%	10%	10%	100%
2	Analyze appropriate research and sampling designs to address business and social science problems.	-	10%	35%	35%	10%	10%	100%
3	Evaluate various stages of research methodology including literature, hypothesis, and data.	-	10%	40%	30%	10%	10%	100%
4	Create a structured research report addressing formulation, ethical concerns, and outcomes.	-	10%	35%	35%	10%	10%	100%

VIII SUGGESTED LEARNING MATERIALS / TEXTBOOKS / REFERENCE BOOKS

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Practical Research Methods	-	Catherine Dawson	How To Books Ltd	2002
2	Research Methodology	2 nd Edition	C. R. Kothari	New Age International	2004
3	Research Methodology: A Step-by-Step Guide	3 rd Edition	Ranjit Kumar	Sage Publications	2011



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Online References

Sr. No	Website Name
A	https://onlinecourses.swayam2.ac.in/ini25_hs10/preview
B	https://onlinecourses.nptel.ac.in/noc23_ge36/preview
C	https://www.youtube.com/watch?v=E2gGF1rburw&list=PLyqSpQzTE6M8F_P8lgjvmqiDEoFGLzG4h
D	https://www.coursera.org/learn/research-methodologies?action=enroll



Program	Final Year B.Tech (Semester VII)	CL	TL	LL	SL	C
Course Name: Disaster Management and Mitigation Measures	Course Code: ETILOCR1705	3	1	-	2	3
Pre-requisite:	NA					

I RATIONALE

A methodical approach to disaster management, encompassing prevention, mitigation, readiness, response, and recovery, is essential for safeguarding people, property, and the environment as well as for creating societies that are safer and more resilient.

II COMPETENCY

Develop the ability to recognize, evaluate, and control the risks associated with a variety of natural and man-made disasters; apply suitable mitigation and prevention techniques; work well with stakeholders; and put into practice workable plans to lessen the effects of disasters, improve community readiness, and guarantee prompt response and recovery in order to create resilient and sustainable communities.

III COURSE OUTCOMES (COs)

After the completion of course based learning students will be able to,

Course Outcome Number	Course Outcome Statement
1	Apply theoretical knowledge to categorize risks, assess Indian disaster scenarios, and its impacts on human life.
2	Analyze the policies, administrative frameworks, modern technologies and public awareness strategies for disaster management in the Indian scenario.
3	Develop integrated financial relief and coordination plans that adhere to legal requirements and successfully involve governmental entities, NGOs and community groups.
4	Create integrated preventive and mitigation strategies covering pre and post disaster planning to enhance community resilience.

IV TEACHING-LEARNING & ASSESSMENT SCHEME

			Learning Scheme						Assessment Scheme														
Sr. No.	Course Title	Course Code	Actual Contact Hrs/Week			SL	Total		Credits		Theory					Practical				SLA		Total	
			CL	TL	LL						Exam	CIA		ESE	Total		CIAP	ESEP	Total				
						Duration in Hrs	MSE					CCE	Max		Min	Max			Min	Max	Min		Max
1	Disaster Management and Mitigation Measures	ETILOCR1705	3	1		2	90	3	2	20	20	60	100	40		Max	Max	Max			25	10	125

V THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
1	CO1	1. Gain a comprehensive understanding of disaster types. 2. Foundational understanding of key disaster management concepts, their relevance to human life. 3. Describe connection between global environmental changes and disaster risks 4. Identify various types of natural disaster 5. Apply knowledge of climate change and global warming to analyze their influence on the frequency and intensity of natural disasters. 6. Apply understanding of various manmade hazards to assess the influence of demographic and societal changes on disaster risks and develop appropriate mitigation and preparedness strategies	Natural Disaster and Manmade Disasters 1.1 Definitions: Definition of Disaster, hazard, global and Indian scenario, general perspective, importance of study in human life, Direct and indirect effects of disasters, long term effects of disasters. Introduction to global warming and climate change. 1.2 Natural Disaster: Meaning and nature of natural disaster, Flood, Flash flood, drought, cloud burst, Earthquake, Landslides, Avalanches, Volcanic eruptions, Mudflow, Cyclone, Storm, Storm Surge, climate change, global warming, sea level rise, ozone depletion. 1.3 Manmade Disasters: Chemical, Industrial, Nuclear and Fire Hazards. Role of growing population and subsequent industrialization, urbanization and changing lifestyle of human beings in frequent occurrences of manmade disasters	12

(Continued on next page)

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
2	CO2	1 Discuss the vital role of disaster management in saving lives, minimizing losses, and strengthening community resilience 2 Analyze the role of effective policies in risk reduction, preparedness enhancement, and timely response and recovery during disasters, and evaluate their impact on overall disaster management efficiency. 3 Analyze the collaborative roles of government agencies, NGOs, and community stakeholders in coordinating response, relief, and recovery efforts during disasters. 4 Interpret a flowchart showing the entire disaster management process, from preparedness and early warning to rescue, relief, and rehabilitation. 5 Analyze the scope, functions, and responsibilities of the National Institute of Disaster Management (NIDM) and the National Disaster Management Authority (NDMA) in India. 6 Analyze comprehensive disaster prevention and mitigation strategies by integrating tools like hazard mapping, early warning mechanisms, modern infrastructure, and community-based preparedness programs. 7 Illustrate with examples how these technologies improve early warning systems, disaster preparedness, response coordination, and post-disaster recovery.	Disaster Management Policy, Framework, and Administration 2.1 Disaster management: meaning, concept, importance, objective of disaster management policy, disaster risks in India, Paradigm shift in disaster management 2.2 Policy and administration: importance and principles of disaster management policies, command and coordination of disaster management, rescue operations—how to start and how to proceed in due course of time, and study of a flowchart showing the entire process. 2.3 Importance of public awareness, preparation, and execution of emergency management programs. Scope and responsibilities of National Institute of Disaster Management (NIDM) and National disaster management authority (NDMA) in India. Methods and measures to avoid disasters, management of casualties, set up of emergency facilities, importance of effective communication among different agencies in such situations. item Modern infrastructure: Use of Internet and software for effective disaster management. Applications of GIS, Remote sensing and GPS in this regard.	12

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
3	CO3	1. Apply appropriate methods to plan and raise financial resources for disaster relief and recovery activities. 2. Demonstrate an understanding of the roles and responsibilities of government agencies and NGOs in mobilizing, managing and distributing financial relief during disasters. 3. Analyze the legal aspects related to fundraising and financial management in disaster scenarios. 4. Identify various active NGOs, evaluate their contributions in past disasters, 5. Develop and implement practical collaboration strategies involving government agencies, NGOs, and community stakeholders to ensure effective disaster response and rehabilitation	Financing Relief Measures 3.1 Role of NGO and government in case of financial Relief Measures: Ways to raise finance for relief expenditure, role of government agencies and NGO's in this process, Legal aspects related to finance raising as well as overall management of disasters. Various NGO's and the works they have carried out in the past on the occurrence of various disasters, Ways to approach these teams.	9

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Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
4	CO4	1. Identify and differentiate between structural and non-structural preventive and mitigation measures for various disaster scenarios. 2. Explain the importance of pre-disaster, during-disaster, and post-disaster strategies. 3. Explain the importance of risk mapping, assessment, and structural defenses such as seawalls, embankments, bio-shields, and shelters. 4. Develop community-based disaster preparedness plans. 5. Evaluate the effectiveness of different measures in reducing disaster risk and enhancing.	Preventive and Mitigation Measures 4.1 Preventive and Mitigation Measures: Pre-disaster, during disaster and post-disaster measures in some events in general Structural mapping: Risk mapping, assessment and analysis, sea walls and embankments, Bio shield, shelters, early warning and communication , Non Structural Mitigation: Community based disaster preparedness, risk transfer and risk financing, capacity development and training, awareness and education, contingency plans.	12

VI SUGGESTED SELF LEARNING ACTIVITIES

1. Create a project proposal for disaster management and mitigation measures (consider location and any disaster; for example, Indian Ocean tsunami, floods, cyclone, landslide, earthquake, etc).
2. Create a mock disaster management plan for any one natural or man-made disaster.
3. Prepare a case study on policy and regulation of disaster management in India.
4. Prepare a case study on structural and non-structural measures in natural disasters.
5. Prepare a case study on the role of NGO in the case of disaster management.
6. Prepare a case study on the role of the United Nations in case of disaster management and mitigation measures.
7. Prepare a case study on the role of the government of India in the case of disaster management.
8. Prepare a case study on modern technology usage in disaster management.
9. Prepare a case study on financial management during disaster management.
10. Prepare a case study on the early warning system and its importance.
11. Explore simulation software for risk modeling (like HAZUS and enQuake) and write a report.
12. Explore role of Artificial Intelligence and machine learning in disaster prediction and response and write a report

VII SUGGESTED WEIGHTAGE TO ASSESSMENT PURPOSE

CO. No.	Course Outcome	BL1	BL2	BL3	BL4	BL5	BL6	Total
1	Apply theoretical knowledge to categorize risks, assess Indian disaster scenarios, and its impacts on human life.	20%	30%	50%	-	-	-	100%
2	Analyze the policies, administrative frameworks, modern technologies and public awareness strategies for disaster management in the Indian scenario.	10%	20%	30%	40%	-	-	100%
3	Develop integrated financial relief and coordination plans that adhere to legal requirements and successfully involve governmental entities, NGOs, and community groups.	10%	20%	30%	20%	10%	10%	100%
4	Create integrated preventive and mitigation strategies covering pre and post disaster planning to enhance community resilience.	10%	20%	30%	20%	10%	10%	100%

VIII SUGGESTED LEARNING MATERIALS / TEXTBOOKS / REFERENCE BOOKS

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Disaster management	-	Harsh Gupta	Universities Press Hill	2003
2	Disaster Management: An Appraisal of Institutional Mechanism in India	-	O S Dagur	published by Centre for land warfare studies	2011
3	Disaster management in India	-	Mr.sunil R Hegde	Notion Press	2021
4	Disaster management in India polices- institutional practices	-	Rajendra K. Pandey	Routledge India	2025

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Natural hazards and disaster management	-	R.B. Singh	Rawat Publications Hill	2006
2	Disaster Management	-	S.C. Sharma	Khanna Publishing House	2018
3	Textbook Of Disaster Management	-	Nitesh Kumar	Satish serial publication	2013

Online References

Sr. No	Website Name
1	https://ndma.gov.in/ , Disaster management authority –Government of in India
2	https://www.undrr.org/ , United nations office for disaster management
3	https://ndma.gov.in/Governance/SDMA , State disaster management authority
4	https://onlinecourses.swayam2.ac.in/cec19_hs20/preview , Online Swayam course on Disaster Management By Naveen Kumar Nanjundan
5	https://nptel.ac.in/courses/105104183 , Nptel online course on Natural Hazards by IIT Kanpur
6	https://onlinecourses.nptel.ac.in/noc25_ar11/preview , Nptel online course on Disaster Recovery and Build Back Better By Prof. Ram Sateesh Pasupuleti, Prof. Subhojyothi Samaddar IIT Roorkee
7	https://elearning.nidm.gov.in/local/home/index.php , E learning on disaster risk reduction



Program	Final Year B. Tech.(Semester VII)	CL	TL	LL	SL	C
Course Name: Entrepreneurship Development and Management	Course Code: ADILOCR1705	4	1	-	3	4
Course Type:	-					
Pre-requisite:	-					

I RATIONALE

This course equips students with essential entrepreneurial knowledge and innovation skills, making them capable of identifying opportunities and building sustainable ventures. It complements their technical background by introducing business thinking, creative problem-solving, and exposure to real-world startup ecosystems. The subject plays a vital role in preparing students for both entrepreneurial and intrapreneurial career paths.

II COMPETENCY

Students will be able to identify real-world problems, generate innovative business ideas, develop viable business models and plans, and evaluate strategies for starting and growing sustainable ventures within a supportive entrepreneurial ecosystem.

III COURSE OUTCOMES (COs)

After the completion of course based learning Students will be able to,

Course Outcome Number	Course Outcome Statement
1	Apply creative problem-solving techniques in entrepreneurial contexts.
2	Develop and present a viable business model and business plan using structured frameworks such as the Business Model Canvas and Pitch Decks.
3	Examine various government policies, institutional frameworks, and ecosystem enablers supporting entrepreneurship and assess their relevance to specific startup contexts.
4	Evaluate entrepreneurial challenges, risk factors, and growth strategies including diversification, acquisition, and sustainability to build resilient businesses.

IV TEACHING-LEARNING & ASSESSMENT SCHEME

				Learning Scheme				Assessment Scheme														
Sr. No.	Course Title	Course Type	Course Code	Actual Contact Hrs/Week			SL	Total	Credits	Exam	Theory				Practical				SLA		Total	
				CL	TL	LL					CIA	ESE	Total		CIAP	ESEP	Total					
													Max	Max			Max	Min	Max	Max		Max
1	Entrepreneurship Development and Management	-	ADILOCR0705	3	1	-	2	90	3	3	20	20	60	100	40	-	-	-	-	-	-	100

V THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
1	CO1	1.1 Explain the fundamentals and functions of entrepreneurship. 1.2 Analyse the role of innovation in business development. 1.3 Apply Design Thinking principles to identify and solve real-world problems. 1.4 Demonstrate creative thinking techniques for entrepreneurial decision-making.	Entrepreneurship, Innovation, and Creative Problem Solving 1.1 Introduction to Entrepreneurship: Definitions by different researchers, Roles, functions, and values of entrepreneurship, Personal traits and qualities of successful entrepreneurs, Case studies of successful entrepreneurs 1.2 Role of Innovation in Entrepreneurship: Concept and importance of innovation, Types of innovation: product, process, business model, Impact of innovation on business growth and competitiveness, National and global innovation-driven startup examples 1.3 Design Thinking for Entrepreneurs: Introduction and principles of Design Thinking, The Design Thinking Process: Empathize, Define, Ideate, Prototype, Test. Real-life applications in startup idea generation, Group activity: Applying Design Thinking to a common local problem 1.4 Creative Thinking Techniques in Entrepreneurship: 1. Divergent Thinking (e.g., brainstorming, SCAMPER) 2. Convergent Thinking (e.g., solution selection and feasibility testing) 3. Lateral Thinking (Edward de Bono techniques) 4. Critical Thinking (structured reasoning for better decisions) 5. Root Cause Analysis using 5 Whys Technique Caselets or short scenarios demonstrating each technique	12

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Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
2	CO2	2.1 Identify and design different types of business models using tools like BMC or Lean Canvas. 2.2 Develop a structured and comprehensive business plan. 2.3 Analyze different strategies for business development including franchising and platform-based models. 2.4 Create and present a basic investor pitch for a startup idea.	Business Models, Planning, and Development Strategies 2.1 Business Model Design: Concept and importance of business models. The Business Model Canvas (BMC): Nine building blocks. Lean Canvas as a startup-friendly alternative. Case studies: Business models of successful startups (e.g., Zomato, Paytm etc). 2.2 Business Plan Formulation: Structure and components of a business plan: • Executive Summary, Company Description, Product/Service, Market Analysis, Marketing & Sales Plan • Organizational & Management Structure, Operational Plan, Financial Plan, Risk Assessment Activity: Drafting a mini business plan in teams 2.3 Business Development Strategies • Forms of business ownership: Sole proprietorship, Partnership, LLP, Private Limited, etc. • Buying an existing business: Advantages, evaluation criteria, and risks • Franchising and Licensing as business growth models • Exploring Digital Platforms and Aggregators (e.g., Swiggy, Airbnb) as new-age business models 2.4 Pitching to Investors • Understanding the structure of an Investor Pitch Deck • Key components: Problem, Solution, Market, Business Model, Traction, Team, Financials • Tips for effective communication and visual storytelling • In-class exercise: Drafting or critiquing a simple pitch deck	12

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Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
3	CO3	3.1 Describe the components and functioning of a startup ecosystem. 3.2 Identify relevant government schemes and institutional frameworks supporting startups. 3.3 Analyze the role of incubators, accelerators, and collaborative platforms in entrepreneurship. 3.4 Evaluate how different ecosystem players contribute to entrepreneurial success.	Entrepreneurial Ecosystem and Institutional Support 3.1 Understanding the Entrepreneurial Ecosystem • Key elements: Entrepreneurs, mentors, investors, service providers, academia, government • Lifecycle needs of startups: idea, seed, early growth, scale-up • Role of incubators, accelerators, startup hubs, and co-working spaces • National initiatives: Startup India, Atal Innovation Mission, NITI Aayog 3.2 Government and Policy Support for Entrepreneurs • MSME and its role in national development • Central & state government schemes and policies: a) PMEGP, MUDRA, Mahila Udyami Yojana, Industrial Cluster Development, CGTMSE b) Women entrepreneurship platforms (WEP), SAMRIDH, Multiplier Grant Scheme 3.3 Institutional Frameworks and Collaborations • AICTE IDEA Lab, IIC (Institution Innovation Council), SAKEC-TBI (customize as per institute) • Role of MoUs and partnerships with industry & government (e.g., Google-GoM MoU) • National Skill Development Mission and its alignment with entrepreneurship • Exposure to angel investors, VCs, startup competitions (e.g., Shark Tank India insights)	8

(Continued on next page)



Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
4	CO4	4.1 Analyze the key operational, financial, and legal challenges in running a business. 4.2 Explain risk and failure in entrepreneurship and strategies to overcome them. 4.3 Evaluate growth strategies and critical factors for scaling a business. 4.4 Appreciate the role of resilience, sustainability, and social impact in entrepreneurship.	Challenges, Growth Strategies, and Entrepreneurial Resilience 4.1 Challenges in Business Development Common issues faced by MSMEs and startups: • Access to finance and markets • Talent acquisition and retention • Regulatory hurdles and compliance (GST, labor laws, etc.) • Supply chain issues and tech infrastructure • Challenges of digital transformation and cybersecurity for small businesses 4.2 Failure and Risk Management • Major causes of startup failure (based on CB Insights reports or India-specific data) • Managing risk and uncertainty: Risk types, risk mitigation strategies • Resilience in entrepreneurship: bouncing back from failure 4.3 Small Business Lifecycle and Growth Strategies • Stages: Seed → Startup → Growth → Maturity → Exit/Revamp • Strategic options: bootstrapping, organic growth, joint ventures, partnerships, franchising Growth Strategies: • Market-Based Growth (Ansoff Matrix) - Market Penetration: Increasing sales in existing markets - Market Development: Entering new geographical or demographic markets - Product Development: Introducing new products to existing markets - Diversification: - Concentric: Related product in new market - Horizontal: Unrelated product in same market - Vertical: Forward or backward integration (e.g., supplier acquisition or direct selling) - Conglomerate: Completely unrelated business lines B. Structural/Strategic Growth - Acquisition: Buying another business for instant access to market or technology - Mergers: Combining businesses to synergize resources and market share - Franchising & Licensing: Rapid expansion with external partners - Strategic Alliances & Joint Ventures: Collaborative growth without full ownership Real-World Case Studies - Zomato–Uber Eats (Acquisition) - Tata Group's diversification - Lenskart's expansion via franchising - Amul's market penetration and product development 4.4 Sustainability and Social Entrepreneurship Principles of sustainable entrepreneurship	10

**VI SUGGESTED SELF LEARNING ACTIVITIES**

CO. No	Activity Title	Description	CO(s)
1	Design Thinking Mini Project	Identify a local problem and use the 5-stage Design Thinking process to propose an innovative business solution.	CO1
2	Business Model Analysis	FAalyze the business model of a successful Indian startup using the Business Model Canvas and suggest one improvement.	CO2
3	Case Studies presentations	Select one successful and one failed business. Thoroughly study them and present your finding in a storytelling form covering introduction, analysis of the business/product/service, factors crucial for the success or failure and key lessons from the cases in the group of 2 to 4 students.	CO3
4	Startup Growth Strategy Poster	Pick a well-known startup and create a visual poster showing its growth strategies (market penetration, acquisitions, etc.) over time.	CO4
5	Investor Pitch Deck Creation	Create a 6–10 slide pitch deck for a startup idea in any domain (tech/non-tech).	CO4

**VII SUGGESTED WEIGHTAGE TO ASSESSMENT PURPOSE**

Sr. No.	Course Outcome	BL1	BL2	BL3	BL4	BL5	BL6	Total
1	Entrepreneurship, Innovation, and Creative Problem Solving	10%	20%	30%	30%	10%	-	100%
2	Apply the different 2D and 3D transformations to demonstrate the effect of different actions on the objects.	-	10%	30%	30%	30%	-	100%
3	Explain the fundamental mechanisms of key human sense organs (e.g., vision, hearing, touch) and articulate their relevance in informing the design principles and challenges of AR-VR systems	10%	60%	30%	-	-	-	100%
4	Apply foundational human-computer interaction (HCI) principles to evaluate the usability and effectiveness of user interfaces in given interactive systems, identifying areas for improvement.	10%	30%	20%	30%	10%	-	100%

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VIII SUGGESTED LEARNING MATERIALS / TEXTBOOKS / REFERENCE BOOKS

Text Books

Sr. No	Title	Edition	Authors	Publisher	Year
1	Entrepreneurship	-	Robert D Hisrich, Michael P Peters, Dean A Shapherd	The McGrawHill Company.	-
2	Entrepreneurial development and management	-	Vasant Desai	Himalaya Publishing House	-
3	Business Environment Text and Cases	30 th Edition	Francis Cherunilam	Himalaya Publishing House	-
4	Karmic Design Thinking - A Buddhism- Inspired Method to Help Create Human-Centered Products & Services	-	Bala Ramdurai	(Self Published)	-
5	Design Thinking: Understand – Improve – Apply	-	Christoph Meinel, Hasso Plattner, Larry Leifer	Springer-Verlag Berlin and Heidelberg GmbH & Co.	-

Reference Books

Sr. No	Title	Edition	Authors	Publisher	Year
1	Marketing Management	17 th Edition	Philip Kotler and Kevin Lane Keller	Pearson Education	-
2	Thinkertoys: A Handbook of Creative-Thinking Techniques	-	Michael Michalko	Ten Speed Press	-
3	Understanding virtual reality: Interface, application and design	2 nd Edition	Alan Craig and William Sherman	Morgan Kaufmann Publisher	2019
4	Fundamentals of Computer Graphics	4 th Edition	Peter Shirley, Michael Ashikhmin, and Steve Marschner	A K Peters/CRC Press	2016



Online References

Sr. No	Website Name
1	https://www.ideou.com
2	https://www.startupindia.gov.in
3	https://www.youtube.com/playlist?list=PL2S3E-gvwhKuxHIX1I9BkQCMcX0m5uDRq
4	https://www.youtube.com/playlist?list=PL5sXpVZS9S0BvBF1hivRN0gjCP-dFteMy
5	https://onlinecourses.swayam2.ac.in/ntr25_ed19/



Program	Final Year B.Tech. Information Technology (Semester VII)	CL	TL	LL	SL	C
Course Name: Secure Application Development Lab	Course Code: ITLR1706	-	-	2	-	1
Pre-requisite:	Programming Language, Web application					

I RATIONALE

The Secure Application Development lab builds software with strong security measures to protect data and systems from threats, while developing essential skills in secure coding critical for modern IT applications.

II COMPETENCY

Develop secure coding skills that meet industry standards and protect applications from real-world cyber threats.

III COURSE OUTCOMES (COs)

After the completion of course based learning Students will be able to,

Course Outcome Number	Course Outcome Statement
1	Apply secure programming techniques by implementing industry best practices and secure coding standards.
2	Classify application vulnerabilities based on OWASP methodologies and standards.
3	Implement robust data validation and authentication mechanisms to ensure input integrity and secure user access.
4	Select appropriate session management techniques to secure communication channels from session-related attacks.

IV TEACHING-LEARNING & ASSESSMENT SCHEME

				Learning Scheme					Assessment Scheme													
Sr. No.	Course Title	Course Type	Course Code	Actual Contact Hrs/Week			SL		Total	Credits	Exam	Theory					Practical				SLA	Total
				CL	TL	LL						MSE	CCE	ESE	Total	CIAP	ESEP	Total				
																			Max	Max		
1	Secure Application Development Lab	VS	ITLR1706	-	-	2	-	30	1	2	-	-	-	-	-	20	30	50	20	-	-	50

V LABORATORY LEARNING OUTCOME AND SUGGESTED LIST OF EXPERIMENTS
 (Minimum 8)

Sr. No.	Laboratory Learning Outcome	Laboratory Experiment Titles	Relevant COs
1	Discuss security laws, compliance requirements and industry standards.	To study different laws and standards of cyber security.	CO2
2	Apply secure coding standards during the SDLC phases with case-based understanding.	Case study for SDLC with secure development practices.	CO1
3	Demonstrate use of source code analysis tools and apply secure version control practices.	To demonstrate SAST tools (Git, GitHub).	CO1
4	Analyze application vulnerabilities based on OWASP Top 10 and implement mitigation strategies.	To study and implement at least any 5 methodologies of OWASP.	CO2
5	Detect command injection vulnerabilities and implement input validation to mitigate them.	Demonstrate OS command injection vulnerability using DVWA.	CO2, CO3
6	Analyze web application vulnerabilities using Burp Suite and interpret the findings.	Use of Burp Suite proxy to test web applications.	CO2
7	Implement frontend and backend validation techniques to ensure secure data input handling.	Apply registration page data validation.	CO1, CO3
8	Identify and prevent SQL injection through secure query techniques and input sanitization.	Apply SQL injection vulnerability that allows login page bypass.	CO2, CO3
9	Apply secure session management practices using tokens, timeouts, and cookie flags.	Study and implement session management for web application.	CO1, CO4
10	Explain and demonstrate the working of symmetric and asymmetric encryption for data confidentiality.	Study types of cryptography – symmetric and asymmetric.	CO1

VI LABORATORY ALIGNED COURSE CONTENT

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
1	CO1	1. Describe cybersecurity laws and regulations. 2. Identify and analyze cyber threats. 3. Study key cybersecurity standards. 4. Apply secure coding methodologies.	Introduction to Secure Programming and Cybersecurity Laws 1.1 Overview of cybersecurity laws, standards, and guidelines. 1.2 Understanding attacks, types of attacks, and statistics on main vulnerabilities. 1.3 Study of different cybersecurity laws and standards. 1.4 Exploration of methodologies for secure code development.	7
2	CO2	1. Describe Secure Software Development Lifecycle (SDLC). 2. Perform Risk Analysis and Threat Modeling. 3. Apply Security Testing Tools.	Methodologies for Developing Secure Code 2.1 Software Development Lifecycle (SDLC) and its phases. 2.2 Risk analysis and threat modeling techniques. 2.3 Examination of Static Application Security Testing (SAST) tools.	8
3	CO3	1. Discuss HTTP Protocol and Web Security Guidelines. 2. Learn Vulnerability Assessment and Penetration Testing (VAPT) Tools. 3. Perform Hands-on Web Application Testing. 4. Analyze and Report Vulnerabilities.	Vulnerability Assessment and Penetration Testing (VAPT) of Applications 3.1 Introduction to HTTP protocol and OWASP Web Security Testing Guidelines. 3.2 Tools for conducting VAPT testing. 3.3 Practical exercises using tools like Burp Proxy for web application testing.	8

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Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
4	CO4	- Discuss Input Data Validation and Authentication Principles. - Analyze Authentication Attacks and Prevention Techniques. - Evaluate and Implement Various Authentication Methods. - Apply Data Validation and Authentication in Practice.	Data Validation, Authentication, and Security Measures 4.1 Guidelines for input data validation and authentication mechanisms. 4.2 Common types of authentication attacks and their prevention measures. 4.3 Study of different types of authentication methods and their implementation. 4.4 Practical exercises focusing on data validation and authentication for login page	7

VII SUGGESTED SELF LEARNING ACTIVITIES

NA

VIII SUGGESTED WEIGHTAGE TO ASSESSMENT PURPOSE

CO No.	Course Outcome	BL1	BL2	BL3	BL4	BL5	BL6	Total
1	Apply secure programming techniques by implementing industry best practices and secure coding standards.	-	20%	80%	-	-	-	100%
2	Classify application vulnerabilities based on OWASP methodologies and standards.	-	-	50%	50%	-	-	100%
3	Implement robust data validation and authentication mechanisms to ensure input integrity and secure user access.	-	20%	80%	-	-	-	100%
4	Select appropriate session management techniques to secure communication channels from session-related attacks.	-	-	40%	50%	10%	-	100%

IX SUGGESTED LEARNING MATERIALS / TEXTBOOKS / REFERENCE BOOKS

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Secure Coding in C and C++	2 nd	Robert C. Seacord	Addison-Wesley Professional	2013

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	The Web Application Hacker's Handbook: Finding and Exploiting Security Flaws	2 nd Edition	Dafydd Stuttard, Marcus Pinto	Wiley	2011
2	Software Security: Building Security In	1 st Edition	Gary McGraw	Addison-Wesley	2006
3	OWASP Testing Guide v4	4 th Edition	OWASP Foundation	OWASP	2014
4	Hacking: The Art of Exploitation	2 nd Edition	Jon Erickson	No Starch Press	2008

Online References

Sr. No	Website Name
A	https://owasp.org , Comprehensive guidelines on web security, OWASP Top 10, Testing Guide, Secure Coding practices.
B	https://www.nist.gov/cyberframework , Official security framework standards from NIST (US).
C	https://attack.mitre.org/k , Repository of known cyberattack techniques, tactics, and procedures.

Program	Final Year B. Tech. Information Technology (Semester VII)	CL	TL	LL	SL	C
Course Name: Major Project-I	Course Code: ITLR1707	-	-	6	-	3
Pre-requisite:	-					

I RATIONALE

This project helps students identify complex engineering problems and develop practical solutions. It provides hands-on experience in managing the entire project lifecycle within a team. The project also strengthens technical, professional, and ethical skills gained during the course and encourages self-learning and research to solve real-world challenges.

II COMPETENCY

By completing this project, students will be competent in analyzing engineering problems, designing viable solutions, managing projects in teams, and demonstrating professionalism and ethical responsibility in real-world scenarios.

III COURSE OUTCOMES (COs)

After the completion of course based learning Students will be able to,

Course Outcome Number	Course Outcome Statement
1	Demonstrate proficiency in identifying, analyzing, and formulating engineering problems related to research/innovation/entrepreneurial within societal and environmental contexts for sustainable development.
2	Comply to the ethical standards and professional codes of conduct in all aspects of their project work, including but not limited to intellectual property rights, privacy concerns, and social responsibility considerations.
3	Design solutions with initial prototypes (hardware/software) for modules of the system as per the defined objectives and scope.
4	Demonstrate the written and oral presentation of the work done effectively.

IV TEACHING-LEARNING & ASSESSMENT SCHEME

			Learning Scheme					Assessment Scheme														
Sr. No.	Course Title	Course Code	Actual Contact Hrs/Week			SL		Total	Credits	Exam	Theory				Practical				SLA		Total	
											Duration in Hrs	CIA		ESE	Total		CIAP	ESEP				Total
			MSE	CCE	ESE	Total		Max				Min	Max		Min	Max						Min
			CL	TL	LL		Notional Hours															
1	Major Project - I	ITLR1707	-	-	6	-	90	3	2	-	-	-	-	-	50	50	100	40	-	-	100	

V GUIDELINES

- Students shall form a group of 3 to 4 students, while forming a group shall not be allowed less than three or more than four students, as it is a group activity
- Students should be informed about the domain and domain experts whose guidance can be taken before selecting projects
- Student's should be recommended to refer papers from reputed conferences/ journals like IEEE, Elsevier, ACM etc. which are not more than 3 years old for review of literature.
- Students can certainly take ideas from anywhere, but be sure that they should evolve them in the unique way to suit their project requirements. Students can be informed to refer to Digital India portal, SIH portal or any other hackathon portal for problem selection.
- Students should do surveys and identify needs, which shall be converted into problem statement for major project in consultation with the Internal guide/Head of department/internal committee of faculties.
- The topic selected should be novel in nature (Product based, Application based or Research based) or should work towards removing the lacuna in currently existing systems.
- The project work can be undertaken in a research institute or organization/Industry/any business establishment. (out-house projects)
- The project proposal presentations can be scheduled according to the domains and should be judged by faculty who are experts in the domain.
- Head of the department and senior staff along with project coordinators will take decisions regarding final selection of projects.
- Students shall submit an implementation plan in the form of Gantt/PERT/CPM chart, which will cover weekly activity of Major Project-I and Major Project-II.
- A log book to be prepared by each group, wherein the group can record weekly work progress, guide can verify, and record notes/comments.
- Internal Guide may give inputs to students during Major Project-I & II activity; however, focus shall be on self-learning.
- Students in a group shall understand problem effectively, propose multiple solutions and select the best possible solution in consultation with the guide.
- Students shall convert the best solution into a working model using various components of their domain areas and demonstrate.
- The solution to be validated with proper justification and report to be compiled in standard format given by the department.
- With the focus on self-learning, innovation, addressing societal problems and entrepreneurship quality development within the students through the Major Project, good quality project need to be carried out in two semesters by all the groups of the students. i.e. Major Project-I in semester VII and Major Project-II in semester VIII.
- Students shall be motivated to publish a paper based on the work in Scopus Conferences/Journals or Copyright or Patent.

VI PROJECT REPORT FORMAT:

At the end of semester VII, each group needs to prepare a project report in Latex as per the guidelines issued by the department.

A project report should preferably contain at least the following details:

- Abstract
- Introduction
- Literature Survey
 - Survey of Existing systems
 - Limitation of Existing systems or research gap
- Problem Statement, Objectives and Scope
- Software Requirement Specification (SRS)
- Project Scheduling and Planning (Use GANTT chart)
- Proposed System
 - Feasibility Study
 - Analysis/Framework/ Algorithm
 - Details of Hardware & Software
 - Design details
 - Methodology (your approach to solve the problem)
- Prototyping and Initial Implementation (Snapshots or code references if available)
- Implementation Plan for Next Semester
 - Timeline Chart for Term-I and Term-II (Project Management tools can be used)
- Conclusion
- Bibliography
- Appendix

VII ASSESSMENT OF MAJOR PROJECT-I PRACTICAL EXAMINATION:

- Report should be prepared as per the guidelines issued by the Department.
- Major Project-I shall be assessed through a presentation and demonstration of the work done by the student project group to a panel of Internal and External Examiners preferably from industry or research organizations having experience of more than five years approved by the head of the department

Suggested quality evaluation parameters are as following:

- Problem Identification
- Literature Survey & Gap Analysis
- Project Proposal & Objectives
- Methodology / Design Approach
- Prototyping / Initial Implementation
- Documentation & Report Quality
- Presentation & Viva
- Team Work

VIII Evaluation Methodology:

- Internal Reviews: 2-3 reviews per semester by guide + departmental committee.
- Final Evaluation: External Examiner + Internal Panel(at end of each semester).

Online References

Sr. No	Website Name
A	https://shahandanchor.knimbus.com/portal/v2/default/home ,KNIMBUS
B	https://link.springer.com/ ,SPRINGER
C	https://ieeexplore.ieee.org/Xplore/home.jsp , IEEE
D	https://sakec.turnitin.com/home/sign-in , TURNITIN
E	https://shorturl.at/cjRpf , B. Tech. LATEX Report Format



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